

Econometrics of Commodity and Asset Pricing

Lecture 6

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- I. Econometrics of Option Pricing
- II. Asset Pricing with Second-Order Esscher Transform

Chapter I

ECONOMETRICS OF OPTION PRICING

– Agenda –

- I.1 Typology of Econometric Asset Pricing Models. (Slide 5)
- I.2 Security Market Models. (Slide 12)
- I.3 Truncated Laplace Transforms (TLT) and Transform Analysis. (Slide 15)
- I.4 Direct Modelling of Conditionally Normal Processes. (Slide 18)
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Chapter I

ECONOMETRICS OF OPTION PRICING

– Agenda –

- ▶ Appendix I.A: Transform Analysis.
- ▶ Appendix I.B: Girsanov in Discrete Time (Cameron Martin Theorem).
- ▶ Appendix I.C: Truncated Laplace Transform of a Normal Distribution.
- ▶ Appendix I.D: Truncated Laplace Transform of a Skewed Laplace Distribution.
- ▶ Appendix I.E: Lemma about ARG Laplace Transform.

I.1 – Typology of Econometric Asset Pricing Models

Definition I.1 (Econometric Asset Pricing Model (EAPM))

An Econometric Asset Pricing Model (EAPM) is defined by the following functions:

- $r_t(\underline{w}_t)$,
- $f(w_{t+1}|\underline{w}_t)$ [or $\psi_t(u)$],
- $\mathcal{M}_{t,t+1}(\underline{w}_{t+1})$,
- $f^{\mathbb{Q}}(w_{t+1}|\underline{w}_t)$ [or $\psi_t^{\mathbb{Q}}(u)$].

- The previous functions have to be specified and parameterized.
They are linked by:

$$f^{\mathbb{Q}}(w_{t+1}|\underline{w}_t) = f(w_{t+1}|\underline{w}_t)\mathcal{M}_{t,t+1}(\underline{w}_{t+1})\exp[r_t(\underline{w}_t)].$$

- If $\mathcal{M}_{t,t+1}$ is exponential-affine:

$$\mathcal{M}_{t,t+1}(\underline{w}_{t+1}) = \exp\{-r_t(\underline{w}_t) + \alpha'_t(\underline{w}_t) w_{t+1} - \psi_t[\alpha_t(\underline{w}_t)]\}.$$

- Once $r_t(\underline{w}_t)$ is specified we have to specify $\psi_t, \alpha_t, \psi_t^{\mathbb{Q}}$ linked by

$$\psi_t^{\mathbb{Q}}(u) = \psi_t(u + \alpha_t) - \psi_t(\alpha_t).$$

We have 3 ingredients (of the EAPM) linked by one (no-arbitrage based) equation: two of them only can be chosen.

- We are going to present three ways of specifying an EAPM:

- 1) the Direct Modelling (from $\mathbb{P}$ and α_t to $\mathbb{Q}$),
- 2) the R.N. Constrained Direct Modelling (or Mixed Modelling),
- 3) the Back Modelling (from $\mathbb{Q}$ and α_t to $\mathbb{P}$).

In all approaches, we have to specify the status of the short rate r_t (developed in next slide).

- Purpose: being $\mathbb{Q}$ -affine (at least) in order to have tractable pricing formulas.

Remark I.1 (The status of the short rate)

The short rate r_t is a function of $\underline{w}_t$, this function may be known or unknown by the econometrician:

i) *It is known in two cases:*

a) *r_t is exogenous $\Rightarrow r_t(\underline{w}_t)$ does not depend on $\underline{w}_t$,*

b) *r_t is an endogenous factor $\Rightarrow r_t$ is a component of $\underline{w}_t$.*

ii) *If the function $r_t(\underline{w}_t)$ is unknown, it has to be specified parametrically:*

$$\{r_t(\underline{w}_t, \tilde{\theta}), \tilde{\theta} \in \tilde{\Theta}\},$$

where $r_t(.,.)$ is a known function.

1) The Direct Modelling

□ Step 1: Specification of the historical dynamics

We choose a parametric family for the conditional historical Log-Laplace transform $\psi_t(u | \underline{w}_t)$:

$$\left\{ \psi_t(u | \underline{w}_t; \theta_{\mathbb{P}}), \theta_{\mathbb{P}} \in \Theta_{\mathbb{P}} \right\}.$$

□ Step 2: Specification of the SDF

$$\mathcal{M}_{t,t+1}(\underline{w}_{t+1}) = \exp \left\{ -r_t(\underline{w}_t, \tilde{\theta}) + \alpha_t(\underline{w}_t)' \underline{w}_{t+1} - \psi_t[\alpha_t(\underline{w}_t) | \underline{w}_t; \theta_{\mathbb{P}}] \right\}.$$

Once r_t has been specified ($r_t(\underline{w}_t, \tilde{\theta})$), $\alpha_t(\underline{w}_t)$ remains to be specified. We assume that $\alpha_t(\underline{w}_t)$ belongs to a parametric family:

$$\left\{ \alpha_t(\underline{w}_t; \theta_{\alpha}), \theta_{\alpha} \in \Theta_{\alpha} \right\}.$$
$$\mathcal{M}_{t,t+1}(\underline{w}_{t+1}, \theta)$$

$$= \exp \left\{ -r_t(\underline{w}_t, \tilde{\theta}) + \alpha_t(\underline{w}_t, \theta_{\alpha})' \underline{w}_{t+1} - \psi_t \left[\alpha_t(\underline{w}_t, \theta_{\alpha}) | \underline{w}_t; \theta_{\mathbb{P}} \right] \right\}, \quad (1.1)$$

where $\theta = (\tilde{\theta}', \theta_{\mathbb{P}}', \theta_{\alpha}')' \in \tilde{\Theta} \times \Theta_{\mathbb{P}} \times \Theta_{\alpha} = \Theta$; $\tilde{\Theta}$ may be reduced to one point [Case i) of Remark 1.1].

1) The Direct Modelling (cont'd)

□ Step 3: Internal consistency conditions (ICC)

For any payoff $g(\underline{w}_s)$ delivered at $s > t$, with price $p(\underline{w}_t)$ at t which is a known function of $\underline{w}_t$, we must have:

$$p_t[g(\underline{w}_s)] = \mathbb{E} \left\{ \mathcal{M}_{t,t+1}(\theta) \dots \mathcal{M}_{s-1,s}(\theta) g(\underline{w}_s) | \underline{w}_t, \theta_{\mathbb{P}} \right\} \forall \underline{w}_t, \theta.$$

These ICC pricing conditions may imply strong constraints on θ , for instance. When a component of $\underline{w}_t$ is a stock return, we have:

$$w_{1,t} = \log \frac{p_{1,t}}{p_{1,t-1}}, \text{ then } \mathbb{E}_t[\mathcal{M}_{t,t+1} \exp(e'_1 \underline{w}_{t+1})] = 1.$$

If this component is a given yield:

$$w_{1,t} = -\frac{1}{h} \log B(t, h), \text{ then } e'_1 \underline{w}_t = -\frac{1}{h} \log \mathbb{E}_t(\mathcal{M}_{t,t+h}).$$

- The previous three steps imply the specification of the R.N. dynamics:

$$\psi_t^{\mathbb{Q}}(u | \underline{w}_t, \theta_{\mathbb{P}}, \theta_{\alpha}) = \psi_t \left[u + \alpha_t(\underline{w}_t, \theta_{\alpha}) | \underline{w}_t, \theta_{\mathbb{P}} \right] - \psi_t \left[\alpha_t(\underline{w}_t, \theta_{\alpha}) | \underline{w}_t, \theta_{\mathbb{P}} \right].$$

- Here, $(\underline{w}_t)$ is chosen to be $\mathbb{P}$ -affine, and the $\mathbb{Q}$ -dynamics is a by-product (it is not chosen ex ante). If the $\mathbb{P}$ -dynamics is Gaussian VAR, and if $r_t(\underline{w}_t, \tilde{\theta})$ and $\alpha_t(\underline{w}_t, \theta_{\alpha})$ are affine functions of $\underline{w}_t$, then the $\mathbb{Q}$ -dynamics remains Gaussian VAR (with different parameters).

2) The R.N. constrained direct modelling (or mixed modelling)

□ Step 1: Specification of $\mathbb{P}$ -dynamics.

We select a family $\{\psi_t(u|\underline{w}_t, \theta_{\mathbb{P}}), \theta_{\mathbb{P}} \in \Theta_{\mathbb{P}}\}$ to explain, for instance, some empirical stylized fact about w_t .

□ Step 2: Specification of $\mathbb{Q}$ -dynamics.

We select a family $\{\psi_t^{\mathbb{Q}}(u|\underline{w}_t, \theta_{\mathbb{Q}}), \theta_{\mathbb{Q}} \in \Theta_{\mathbb{Q}}\}$ of affine $\mathbb{Q}$ -dynamics, and possibly $\{r_t(\underline{w}_t, \tilde{\theta}), \tilde{\theta} \in \tilde{\Theta}\}$, to have explicit or quasi explicit pricing formula.

- We may want to choose $\mathbb{P}$ - and $\mathbb{Q}$ -dynamics such that, for any pair $(\psi_t, \psi_t^{\mathbb{Q}})$ belonging to these families, there exists a unique function $\alpha_t(\underline{w}_t, \theta_{\mathbb{P}}, \theta_{\mathbb{Q}})$ characterizing the exponential-affine SDF such that:

$$\psi_t^{\mathbb{Q}}(u|\underline{w}_t, \theta_{\mathbb{Q}}) = \psi_t \left[u + \alpha_t(\underline{w}_t, \theta_{\mathbb{P}}, \theta_{\mathbb{Q}}) | \underline{w}_t, \theta_{\mathbb{P}} \right] - \psi_t \left[\alpha_t(\underline{w}_t, \theta_{\mathbb{P}}, \theta_{\mathbb{Q}}) | \underline{w}_t, \theta_{\mathbb{P}} \right]. \quad (1.2)$$

This condition is generally satisfied only for a subset of pairs $(\theta_{\mathbb{P}}, \theta_{\mathbb{Q}})$.
The SDF is anyway a by-product.

□ Step 3: Internal Consistency Conditions (ICC).

Once the (constrained) parameterization $(\tilde{\theta}, \theta_{\mathbb{P}}, \theta_{\mathbb{Q}}) \in \tilde{\Theta} \times \Theta_{\mathbb{P}} \times \Theta_{\mathbb{Q}}$ is formalized, internal consistency conditions may be imposed.

- If the unique function $\alpha_t(\underline{w}_t)$ that satisfies (1.2) does not exist, we can still adopt this mixed modeling strategy but in that case, the SDF is not exponential-affine.

3) The Back Modelling

- **Step 1: Specification of the R.N. dynamics** [and possibly of $\{r_t(\underline{w}_t, \tilde{\theta}), \tilde{\theta} \in \tilde{\Theta}\}$]: $\{\psi_t^Q(u|\underline{w}_t; \theta_Q), \theta_Q \in \Theta_Q\}$.
- **Step 2: Internal consistency conditions (ICC)**, if relevant (and $\forall \underline{w}_t, \tilde{\theta}, \theta_Q$):

$$p_t[g(\underline{w}_s)] = \mathbb{E}_t^Q \left[\exp(-r_t(\underline{w}_t, \tilde{\theta}) - \dots - r_{s-1}(\underline{w}_s, \tilde{\theta})) g(\underline{w}_s) | \underline{w}_t, \theta_Q \right].$$

If $w_{1,t} = \log(p_{1,t}/p_{1,t-1})$, its ICC is $r_t = \psi_t^Q(e_1 | \underline{w}_t; \theta_Q)$.

- **Step 3: Choice of the specification of $\alpha_t(\underline{w}_t)$** (without any constraint), providing the family $\{\alpha_t(\underline{w}_t, \theta_\alpha), \theta_\alpha \in \Theta_\alpha\}$. The $\mathbb{P}$ -dynamics is a by-product:

$$\psi_t(u|\underline{w}_t; \theta_Q, \theta_\alpha) = \psi_t^Q \left[u - \alpha_t(\underline{w}_t, \theta_\alpha) | \underline{w}_t; \theta_Q \right] - \psi_t^Q \left[-\alpha_t(\underline{w}_t, \theta_\alpha) | \underline{w}_t, \theta_Q \right].$$

- If the R.N. conditional p.d.f. $f_t^Q(w_{t+1}|\underline{w}_t, \theta_1^*)$ is known in (quasi) closed form, the same is true for the historical conditional p.d.f.:

$$f_t(w_{t+1}|\underline{w}_t, \theta_Q, \theta_\alpha) = f_t^Q(w_{t+1}|\underline{w}_t, \theta_Q) \exp \left\{ -\alpha_t'(\underline{w}_t, \theta_\alpha) w_{t+1} - \psi_t^Q \left[-\alpha_t(\underline{w}_t, \theta_\alpha) | \underline{w}_t, \theta_Q \right] \right\},$$

- Here, (w_t) is chosen to be $\mathbb{Q}$ -affine, and the $\mathbb{P}$ -dynamics is a by-product (it is not chosen ex ante).

I.2 – Security Market Models

The setup

- In this chapter we assume $w_t = (y_t', z_t')'$, where:
 - y_t denotes a K_1 -dimensional vector of (observable) geometric stock returns;
 - z_t is a K_2 -dimensional vector of variables not observed by the econometrician;
 - w_t is the K -dimensional state vector, with $K = K_1 + K_2$.
- The short rate process $r_t = -\ln B(t, 1)$ is exogenous (i.e. constant function of $\underline{w}_t$):
 - it is the one-period geometric return of the short-term risk-free ZCB; we thus work conditionally to this process r_t .
- We want to determine the date- t no-arbitrage price of a derivative asset with payoff $g(\underline{y}_T, \underline{z}_T)$ at $T > t$. Let us consider in particular the payoff of a European Call option maturing in T , with strike price K and written on $p_{1,t}$:

$$g(\underline{y}_T, \underline{z}_T) = (p_{1,T} - K)^+ = p_{1,t} \left[\exp(y_{1,t+1} + \dots + y_{1,T}) - \kappa_1 \right]^+,$$

where $\kappa_1 = K/p_{1,t}$.

I.2 – Security Market Models

Internal Consistency Conditions (ICC)

- We know that the no-arbitrage price at date t of $g(\underline{y}_T, \underline{z}_T)$ is:

$$\begin{aligned} p_t &= \mathbb{E}_t[\mathcal{M}_{t,t+1} \dots \mathcal{M}_{T-1,T} g(\underline{y}_T, \underline{z}_T)] \\ &= \exp(-r_t \dots - r_{T-1}) \mathbb{E}_t^{\mathbb{Q}}[g(\underline{y}_T, \underline{z}_T)]. \end{aligned}$$

- For each $j \in \{1, \dots, K_1\}$ we have:

$$1 = \mathbb{E}_t[\mathcal{M}_{t,t+1} \exp(y_{j,t+1})],$$

and therefore we have prices at t which are known functions of $\underline{w}_t$, namely 1; $\Rightarrow$ we have thus to guarantee ICC.

- In the Direct Modelling approach, since:

$$\mathcal{M}_{t,t+1} = \exp\left(-r_t + \alpha'_t(\underline{w}_t; \theta_\alpha) \underline{w}_{t+1} - \psi_t[\alpha_t(\underline{w}_t; \theta_\alpha) \mid \underline{w}_t, \theta_{\mathbb{P}}]\right)$$

these conditions are:

$$r_t = \psi_t[\alpha_t(\underline{w}_t; \theta_\alpha) + e_j \mid \underline{w}_t, \theta_{\mathbb{P}}] - \psi_t[\alpha_t(\underline{w}_t; \theta_\alpha) \mid \underline{w}_t, \theta_{\mathbb{P}}],$$

$$\forall \underline{w}_t, \theta_{\mathbb{P}}, \theta_\alpha; j \in \{1, \dots, K_1\}.$$

I.2 – Security Market Models

Internal Consistency Conditions (ICC)

- In the **Back Modelling** (and in the Risk-Neutral Constrained Direct Modelling approach) these conditions are :

$$r_t = \psi_t^{\mathbb{Q}}(e_j | \underline{w}_t; \theta_{\mathbb{Q}}), \forall \underline{w}_t, \theta_{\mathbb{Q}}; j \in \{1, \dots, K_1\}.$$

- Note that, these conditions can also be equivalently written as:

- $\mathbb{E}_t^{\mathbb{Q}} \exp(y_{j,t+1}) = \exp(r_t), j = 1, \dots, K_1;$
- or $\mathbb{E}_t^{\mathbb{Q}}(\rho_{A,j,t+1} - r_{A,t+1}) = 0$, that is the excess arithmetic returns are $\mathbb{Q}$ -martingale differences.

- If (w_t) is $\mathbb{Q}$ -affine, with conditional risk neutral Log-Laplace transform $\psi_t^{\mathbb{Q}}(u | \underline{w}_t) = a^{\mathbb{Q}}(u)' w_t + b_t^{\mathbb{Q}}(u)$, the previous **internal consistency conditions** are given by (using the Back Modelling notation) :

$$\begin{cases} a^{\mathbb{Q}}(e_j, \theta_{\mathbb{Q}}) = 0 \\ b_t^{\mathbb{Q}}(e_j, \theta_{\mathbb{Q}}) = r_t, \forall \theta_{\mathbb{Q}}; j \in \{1, \dots, K_1\}. \end{cases}$$

I.3 – Truncated Laplace Transforms and Transform Analysis

- Let us introduce the following notation:

$$w_{t+1,T} = (w'_{t+1}, w'_{t+2}, \dots, w'_T)'$$

with w_t affine K -dimensional process.

- We want to compute:

$$\tilde{\varphi}_t(u; v, \gamma) = \mathbb{E}_t[\exp(u' w_{t+1,T}) \mathbb{1}_{\{v' w_{t+1,T} < \gamma\}}].$$

- Consider the complex untruncated conditional Laplace transform:

$$\varphi_t(z) = \mathbb{E}_t[\exp(z' w_{t+1,T})], \quad z \in \mathbb{C}^{K \times T},$$

computed using the same recursive algorithm as in the real case.

I.3 – Truncated Laplace Transforms and Transform Analysis

- We have [see Duffie, Pan, Singleton (2000) and Appendix I.A]:

$$\tilde{\varphi}_t(u; v, \gamma) = \frac{\varphi_t(u)}{2} - \frac{1}{\pi} \int_0^\infty \frac{\text{Im}[\varphi_t(u + ivx) \exp(-i\gamma x)]}{x} dx. \quad (I.3)$$

where Im means imaginary part. Note that the integral in Eq. (I.3) is one dimensional (whatever the dimension of w_t).

- In option pricing, the typical problem is to compute a conditional expectation of the type (with $k > 0$):

$$\begin{aligned} & \mathbb{E}_t \left([\exp(u'_1 w_{t+1, T}) - k \exp(u'_2 w_{t+1, T})]^+ \right) \\ &= \mathbb{E}_t \left([\exp(u'_1 w_{t+1, T}) - k \exp(u'_2 w_{t+1, T})] \mathbb{1}_{\{[\exp(u_1 - u_2)' w_{t+1, T}] > k\}} \right) \\ &= \tilde{\varphi}_t(u_1; u_2 - u_1, -\log k) - k \tilde{\varphi}_t(u_2; u_2 - u_1, -\log k). \end{aligned}$$

I.3 – Truncated Laplace Transforms and Transform Analysis

Example I.1 (Exogenous short rate)

- Consider an asset whose date- t price is p_t . We have a deterministic (exogenous) risk-free rate r_t .
- One-period geometric stock return: $y_{1,t+1} = \log(p_{1,t+1}/p_{1,t})$.
- Consider a European Call option written on this asset, maturing in $t + h$ and with strike price K .
- Payoff: $(p_{1,t+h} - K)^+ = p_{1,t} [\exp(y_{1,t+1} + \dots + y_{1,t+h}) - \kappa_1]^+$, with $\kappa_1 = K/p_{1,t}$.
- The option price is :

$$p_{1,t} \exp(-r_t - \dots - r_{t+h-1}) \mathbb{E}_t^{\mathbb{Q}} \left\{ [\exp u_1' w_{t+1,t+h} - \kappa_1]^+ \right\},$$

with $u_1 = e \otimes e_1$, where e is the h -dimensional vector with components 1 and e_1 the K -vector selecting the 1st component ($y_{1,t}$ is the 1st component of y_t and w_t).

I.4 – Direct Modelling of Conditionally Normal Processes

$\mathbb{P}$ -dynamics, SDF and $\mathbb{Q}$ -dynamics

- The short rate r_t is exogenous and $w_t = y_t$ is given by a vector of observed geometric stock returns.
- The $\mathbb{P}$ -dynamics of (y_t) is given by:

$$y_{t+1} = m(\underline{y}_t, \theta_{\mathbb{P}}) + \Sigma^{1/2}(\underline{y}_t, \theta_{\mathbb{P}}) \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \text{IIN}(0, Id_K)$$

with possible specifications given by ARMA, GARCH, ARMA-GARCH, GARCH-M and Threshold-GARCH processes.

- Given that the ICC are given by:

$$r_t = \psi_t(\alpha_t + e_j) - \psi_t(\alpha_t), j = 1, \dots, K;$$

with $\psi_t(u) = u' m_t + \frac{1}{2} u' \Sigma_t u$, then we have (with e_j the j^{th} vector of the canonical basis):

$$r_t = e_j' m_t + e_j' \Sigma_t \alpha_t + \frac{1}{2} \Sigma_{jj,t}$$

or

$$r_t e = m_t + \Sigma_t \alpha_t + \frac{1}{2} \text{vdiag}(\Sigma_t),$$

I.4 – Direct Modelling of Conditionally Normal Processes

$\mathbb{P}$ -dynamics, SDF and $\mathbb{Q}$ -dynamics

□ We find:

$$\alpha_t = \Sigma_t^{-1} \left(r_t e - m_t - \frac{1}{2} v \operatorname{diag} \Sigma_t \right)$$

which is a known function of historical parameters only (no $\theta_{\mathbb{Q}}$). The risk-neutral conditional log-Laplace transform is given by:

$$\begin{aligned} \psi_t^{\mathbb{Q}}(u) &= \psi_t(u + \alpha_t) - \psi_t(\alpha_t) \\ &= (u + \alpha_t)' m_t + \frac{1}{2} (u + \alpha_t)' \Sigma_t (u + \alpha_t) - \alpha_t' m_t - \frac{1}{2} \alpha_t' \Sigma_t \alpha_t \\ &= u' (m_t + \Sigma_t \alpha_t) + \frac{1}{2} u' \Sigma_t u \\ &= u' [r_t e - \frac{1}{2} v \operatorname{diag}(\Sigma_t)] + \frac{1}{2} u' \Sigma_t u, \end{aligned}$$

and therefore the $\mathbb{Q}$ -dynamics is given by:

$$y_{t+1} = r_t e - \frac{1}{2} v \operatorname{diag}(\Sigma_t) + \Sigma_t^{1/2} \varepsilon_{t+1}^{\mathbb{Q}}, \quad \varepsilon_{t+1}^{\mathbb{Q}} \stackrel{\mathbb{Q}}{\sim} \operatorname{IIN}(0, Id_K).$$

□ We also have:

$$\begin{aligned} \varepsilon_{t+1} &= \Sigma_t^{-1/2} (r_t e - m_t - 1/2 \operatorname{diag} \Sigma_t) + \varepsilon_{t+1}^{\mathbb{Q}} \\ &= \lambda_t + \varepsilon_{t+1}^{\mathbb{Q}} \text{ (say)}. \end{aligned}$$

I.4 – Direct Modelling of Conditionally Normal Processes

$\mathbb{P}$ -dynamics, SDF and $\mathbb{Q}$ -dynamics

- We could present this result, in an equivalent way (similar to the continuous time framework), starting from the SDF:

$$\begin{aligned}\mathcal{M}_{t,t+1} &= \exp \left(\alpha'_t y_{t+1} - r_t - \alpha'_t m_t - \frac{1}{2} \alpha'_t \Sigma_t \alpha_t \right) \\ &= \exp \left(-r_t + \alpha'_t \Sigma_t^{1/2} \varepsilon_{t+1} - \frac{1}{2} \alpha'_t \Sigma_t \alpha_t \right),\end{aligned}$$

- and the associated change of probability measure:

$$\begin{aligned}d_{t+1}^{\mathbb{Q}} &= \exp(\alpha'_t y_{t+1} - \psi_t(\alpha_t)) \\ &= \exp \left(\alpha'_t \Sigma_t^{1/2} \varepsilon_{t+1} - \frac{1}{2} \alpha'_t \Sigma_t \alpha_t \right) \\ &= \exp \left(\lambda'_t \varepsilon_{t+1} - \frac{1}{2} \lambda'_t \lambda_t \right)\end{aligned}$$

$$\text{with } \lambda_t = \Sigma_t^{1/2} \alpha_t = \Sigma_t^{-1/2} \left(r_t e - m_t - \frac{1}{2} v \text{diag} \Sigma_t \right).$$

I.4 – Direct Modelling of Conditionally Normal Processes

$\mathbb{P}$ -dynamics, SDF and $\mathbb{Q}$ -dynamics

- From the Girsanov (Cameron Martin) theorem (since the previous function is also the p.d.f. of the R.N. conditional distribution of ε_{t+1} w.r. to the corresponding historical distribution, $N(0, Id_K)$; see Appendix I.B) we have:

$$\varepsilon_{t+1} = \lambda_t + \varepsilon_{t+1}^{\mathbb{Q}}, \quad \varepsilon_{t+1}^{\mathbb{Q}} \stackrel{\mathbb{Q}}{\sim} IIN(0, Id_K)$$

and therefore:

$$\begin{aligned} y_{t+1} &= m_t + \Sigma_t^{1/2} (\lambda_t + \varepsilon_{t+1}^{\mathbb{Q}}) \\ &= r_t e - \frac{1}{2} v \operatorname{diag} \Sigma_t + \Sigma_t^{1/2} \varepsilon_{t+1}^{\mathbb{Q}} \end{aligned}$$

- So, we have the following **key result**:

$$f^{\mathbb{Q}}(y_{t+1} | \underline{y}_t, \theta_{\mathbb{P}}) = n \left(y_{t+1}; r_{t+1} e - \frac{1}{2} v \operatorname{diag} \Sigma_t, \Sigma_t \right)$$

and

$$y_{t+1} = r_t e - \frac{1}{2} v \operatorname{diag} \Sigma_t + \Sigma_t^{1/2} \varepsilon_{t+1}^{\mathbb{Q}}, \quad \varepsilon_t^{\mathbb{Q}} \stackrel{\mathbb{Q}}{\sim} IIN(0, Id_K)$$

I.4 – Direct Modelling of Conditionally Normal Processes

Pricing

- The date- t price of a European Call option with unitary residual maturity, written on the first stock return and with strike price $\kappa = \kappa_1$ ($p_{1,t} = 1$) is:

$$C_t = \exp(-r_t) \mathbb{E}_t^Q \left\{ [\exp(y_{1,t+1}) - \kappa]^+ \right\}$$

- The risk-neutral conditional distribution of $y_{1,t+1}$ given the past is $N(r_t - \frac{1}{2}\sigma_t^2, \sigma_t^2)$.

- It is easily seen (see Appendix I.C) that if $X \sim N(0, 1)$, its truncated Laplace transform is, for $v < 0$:

$$\begin{aligned} \varphi_X(u, v, \gamma) &= \mathbb{E}[\exp(uX) \mathbb{1}_{\{vX < \gamma\}}] \\ &= \exp\left(\frac{u^2}{2}\right) \Phi\left(-\frac{\gamma}{v} + u\right) \end{aligned}$$

I.4 – Direct Modelling of Conditionally Normal Processes

Pricing

□ Therefore:

$$\begin{aligned}C_t &= \exp(-r_t) \mathbb{E}_t^{\mathbb{Q}}[\exp(y_{t+1}) \mathbb{1}_{(y_{t+1} > \log \kappa)} - \kappa \mathbb{1}_{(y_{t+1} > \log \kappa)}] \\&= \exp(-r_t) \mathbb{E}_t^{\mathbb{Q}} \left[\exp \left(r_t - \frac{1}{2} \sigma_t^2 + \sigma_t \varepsilon_{t+1}^{\mathbb{Q}} \right) \mathbb{1}_{(\sigma_t \varepsilon_{t+1}^{\mathbb{Q}} > \log \kappa - r_t + (1/2) \sigma_t^2)} \right. \\&\quad \left. - \kappa \mathbb{1}_{(\sigma_t \varepsilon_{t+1}^{\mathbb{Q}} > \log \kappa - r_t + (1/2) \sigma_t^2)} \right] \\&= \exp \left(-\frac{1}{2} \sigma_t^2 \right) \varphi_{\varepsilon}^{\mathbb{Q}} \left(\sigma_t, -\sigma_t, -\log \kappa + r_t - \frac{1}{2} \sigma_t^2 \right) \\&\quad - \kappa \exp(-r_t) \varphi_{\varepsilon}^{\mathbb{Q}} \left(0, -\sigma_t, -\log \kappa + r_t - \frac{1}{2} \sigma_t^2 \right) \\&= \Phi \left[\frac{r_t - \log \kappa - \frac{1}{2} \sigma_t^2}{\sigma_t} + \sigma_t \right] \\&\quad - \kappa \exp(-r_t) \Phi \left[\frac{r_t - \log \kappa - \frac{1}{2} \sigma_t^2}{\sigma_t} \right]\end{aligned}$$

I.4 – Direct Modelling of Conditionally Normal Processes Pricing

□ Key result:

$$C_t = \Phi \left(\frac{r_t - \log \kappa}{\sigma_t} + \frac{1}{2} \sigma_t \right) - \kappa \exp(-r_t) \Phi \left(\frac{r_t - \log \kappa}{\sigma_t} - \frac{1}{2} \sigma_t \right) .$$

which is the well-known Black-Scholes formula (for a price of the underlying asset equal to 1 and maturity 1).

- ▶ For a residual maturity $T - t$ larger than one, we need the R.N. conditional joint Laplace transform of $\tilde{w}_{t+1, T}$.
- The (conditional) joint RN distribution is Gaussian if and only if Σ_t is non random, i.e. if y_{t+1} is, in the RN world, an independent gaussian process of marginal distribution $N \left(r_t - \frac{1}{2} v \text{diag} \Sigma_t, \Sigma_t \right)$, Σ_t non random.

I.4 – Direct Modelling of Conditionally Normal Processes

Pricing

- In this particular case, we get a Black-Scholes pricing formula for the maturity T .
- We replace r_t by $(r_t + \dots + r_{T-1})$ and σ_t by $\sqrt{(T-t)}\sigma$, if the variances are identical or by $(\sigma_t^2 + \dots + \sigma_{T-1}^2)^{1/2}$ otherwise.
- If Σ_t does depend on $\underline{y}_t$ we can easily compute $C_{t,T}$ by Monte Carlo.

key result:

$$C_{t,T} \simeq \frac{1}{S} \exp(-r_t - \dots - r_{T-1}) \sum_{s=1}^S [\exp(y_{1,t+1}^s + \dots + y_{1,T}^s) - k]^+,$$

where $(y_{t+1}^s, \dots, y_T^s)$, $s = 1, \dots, S$, are S paths drawn independently and recursively in the one-step conditional multivariate Gaussian distributions.

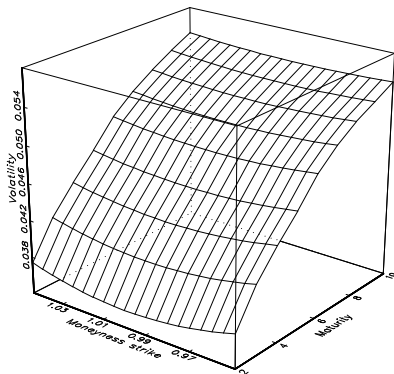
- key result: $m(y_t)$ does not play any role.
- Next slides: smiles generated by nonlinear (ARCH-like) dynamics, in spite of conditional normality.

I.4 – Direct Modelling of Conditionally Normal Processes

Figures

- Figures 1 [$r_t = 0$ and $\sigma_{t+1}^2 = 0.001 + 0.95y_t^2$ (ARCH(1) model)];

FIGURE 1: Implied volatility, ARCH model

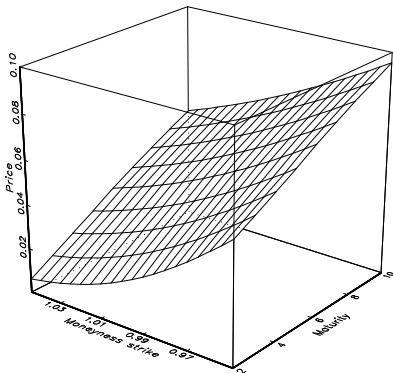


I.4 – Direct Modelling of Conditionally Normal Processes

Figures

- Figure 2 [$r_t = 0$ and $\sigma_{t+1}^2 = 0.001 + 0,95y_t^2$ (ARCH(1) model)];

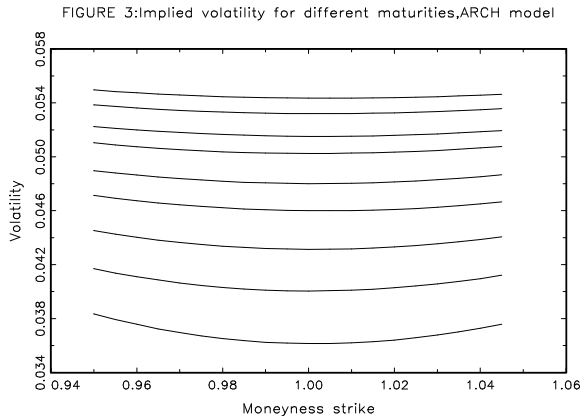
FIGURE 2: Call price,ARCH model



I.4 – Direct Modelling of Conditionally Normal Processes

Figures

- Figure 3 [$r_t = 0$ and $\sigma_{t+1}^2 = 0.001 + 0.95y_t^2$ (ARCH(1) model)];

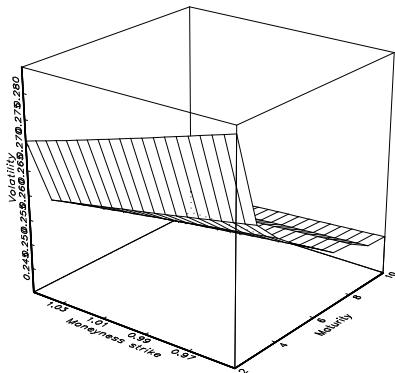


I.4 – Direct Modelling of Conditionally Normal Processes

Figures

- Figure 4 [$r_t = 0$ and $\sigma_{t+1}^2 = 0.1 \mathbb{1}_{\{y_t < 0\}} + 0.001 \mathbb{1}_{\{y_t > 0\}}$ (Threshold ARCH(1) model)];

FIGURE 4: Implied Volatility, piecewise linear conditional variance

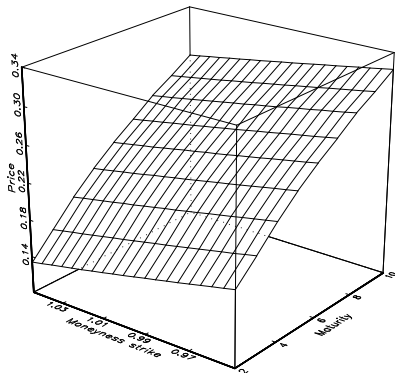


I.4 – Direct Modelling of Conditionally Normal Processes

Figures

- Figure 5 [$r_t = 0$ and $\sigma_{t+1}^2 = 0.1 \mathbb{1}_{\{y_t < 0\}} + 0.001 \mathbb{1}_{\{y_t > 0\}}$ (Threshold ARCH(1) model)];

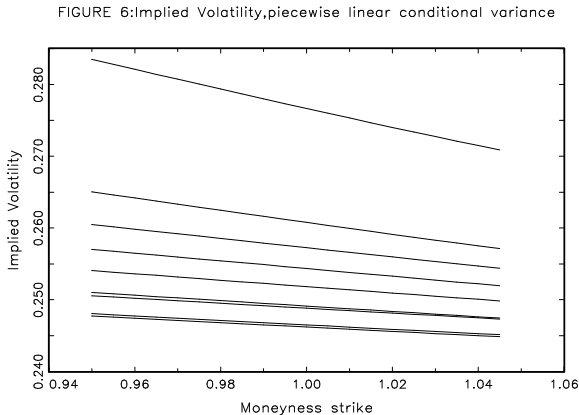
FIGURE 5: Call price, piecewise linear conditional variance



I.4 – Direct Modelling of Conditionally Normal Processes

Figures

- Figure 6 [$r_t = 0$ and $\sigma_{t+1}^2 = 0.1 \mathbb{1}_{\{y_t < 0\}} + 0.001 \mathbb{1}_{\{y_t > 0\}}$ (Threshold ARCH(1) model)];



I.5 – Back Modelling of Conditionally Normal Processes

Q-dynamics, SDF and $\mathbb{P}$ -dynamics

i) In this case we start with the risk neutral dynamics:

$$\psi_t^{\mathbb{Q}}(u \mid \underline{y}_t, \theta_{\mathbb{Q}}) = u' m_t^{\mathbb{Q}}(\underline{y}_t, \theta_{\mathbb{Q}}) + \frac{1}{2} u' \Sigma_t^{\mathbb{Q}}(\underline{y}_t, \theta_{\mathbb{Q}}) u$$

ii) We impose the ICC:

$$r_t = m_{jt}^{\mathbb{Q}}(\underline{y}_t, \theta_{\mathbb{Q}}) + \frac{1}{2} \Sigma_{jj,t}^{\mathbb{Q}}, j \in \{1, \dots, K\},$$

and the R.N. dynamics compatible with the A.A.O. is :

$$\psi_t^{\mathbb{Q}}(u \mid \underline{y}_t, \theta_{\mathbb{Q}}) = u' e r_t - \frac{1}{2} u' v \text{diag} \Sigma_t^{\mathbb{Q}} + \frac{1}{2} u' \Sigma_t^{\mathbb{Q}} u,$$

that is:

$$y_{t+1} = r_t e - \frac{1}{2} v \text{diag}(\Sigma_t^{\mathbb{Q}}) + (\Sigma_t^{\mathbb{Q}})^{1/2} \varepsilon_{t+1}^{\mathbb{Q}}.$$

I.5 – Back Modelling of Conditionally Normal Processes

Q-dynamics, SDF and P-dynamics

iii) Choosing any $\alpha_t(\underline{y}_t, \theta_\alpha)$, we deduce the historical dynamics:

$$\begin{aligned}\psi_t(u \mid \underline{y}_t; \theta_Q, \theta_\alpha) \\&= \psi_t^Q(u - \alpha_t) - \psi_t^Q(-\alpha_t) \\&= u' \left[e r_t - \frac{1}{2} \text{vdiag} \Sigma_t^Q(\underline{y}_t; \theta_Q) \right. \\&\quad \left. - \Sigma_t^Q(\underline{y}_t; \theta_Q) \alpha_t(\underline{y}_t, \theta_\alpha) \right] + \frac{1}{2} u' \Sigma_t^Q(\underline{y}_t; \theta_Q) u,\end{aligned}$$

and θ_Q and θ_α can be identified from the dynamics of y_t only. The P-dynamics is given by:

$$y_{t+1} = r_t e - \frac{1}{2} \text{vdiag}(\Sigma_t^Q) - \Sigma_t^Q \alpha_t + (\Sigma_t^Q)^{1/2} \varepsilon_{t+1}.$$

- The Q-dynamics is discrete-time affine if $\Sigma_t^Q = \Sigma^Q$.
- Since α_t arbitrary, any historical conditional mean can be reached, but we have $\Sigma_t = \Sigma^Q$.

I.6 – Direct Modelling of Mixed-Normal Processes

Static case

- Based on Bertholon, Monfort and Pegoraro (2006).
- Motivation: the skewness $\tilde{\mu}_3$ and kurtosis $\tilde{\mu}_4$ of a random variable with any probability distribution span the domain:

$$\mathcal{D} = \{(\tilde{\mu}_3, \tilde{\mu}_4) \in \mathbb{R} \times \mathbb{R}_+^* : \tilde{\mu}_4 \geq \tilde{\mu}_3^2 + 1\}.$$

- In the $(\tilde{\mu}_3, \tilde{\mu}_4)$ -plane, the boundary of the $(\tilde{\mu}_3, \tilde{\mu}_4)$ domain of maximal size $\mathcal{D}$ is a parabola in which $\tilde{\mu}_4$ is bounded from below by 1.
- It is easily seen that any possible pair of $(\tilde{\mu}_3, \tilde{\mu}_4)$ in the maximal set $\mathcal{D}$ can be reached by a mixture of only two Normal distributions.
- See Bertholon, Monfort and Pegoraro (2006) for further details.

I.6 – Direct Modelling of Mixed-Normal Processes

Static case

- Let us consider $K = 1$ and $w_t = y_t$ with $y_{t+1} \sim \text{IMN}(J, p_j, \mu_j, \sigma_j^2)$; the historical p.d.f. is given by:

$$f(y_{t+1}) = \sum_{j=1}^J p_j n(y_{t+1}; \mu_j, \sigma_j^2),$$

with an associated Laplace transform given by:

$$\varphi(u) = \sum_{j=1}^J p_j \exp\left(u\mu_j + \frac{1}{2}u^2\sigma_j^2\right).$$

- The one-period positive SDF is given by:

$$\mathcal{M}_{t,t+1} = \exp(\alpha_t y_{t+1} + \beta_t) = \frac{\exp(-r_t + \alpha_t y_{t+1})}{\varphi(\alpha_t)}$$

I.6 – Direct Modelling of Mixed-Normal Processes

Static case

- The ICC $\varphi(\alpha_t + 1) = \exp(r_t) \varphi(\alpha_t)$ has a unique solution in α_t . The risk neutral p.d.f. is given by:

$$f^{\mathbb{Q}}(y_{t+1}) = \sum_{j=1}^J \nu_{jt} n(y_{t+1}; \mu_j + \alpha_t \sigma_j^2, \sigma_j^2)$$

$$\text{with } \nu_{jt} = \frac{p_j \exp\left(\alpha_t \mu_j + \frac{1}{2} \alpha_t^2 \sigma_j^2\right)}{\sum_{j=1}^J p_j \exp\left(\alpha_t \mu_j + \frac{1}{2} \alpha_t^2 \sigma_j^2\right)}.$$

- We thus have a risk neutral mixed-normal distribution with Laplace transform:

$$\varphi_t^{\mathbb{Q}}(u) = \sum_{j=1}^J \nu_{jt} \exp\left[u\left(\mu_j + \alpha_t \sigma_j^2\right) + \frac{1}{2} \sigma_j^2 u^2\right].$$

It is a linear combination of Gaussian Laplace transforms and therefore we have explicit computation of the truncated Laplace transform.

I.6 – Direct Modelling of Mixed-Normal Processes

Static case

- More precisely, the price at t of the payoff $[\exp(y_{t+1} - \kappa)]^+$ is given by:

$$C_t = \sum_{j=1}^J \nu_{jt} \gamma_{jt} C_{BS} \left(\sigma_j^2, \frac{\kappa}{\gamma_{jt}} \right)$$

with

$$\gamma_{jt} = \exp \left(\mu_j + \alpha_t \sigma_j^2 - r_t + \frac{\sigma_j^2}{2} \right), \quad 0 \leq \nu_{jt} \gamma_{jt} \leq 1, \quad \sum_{j=1}^J \nu_{jt} \gamma_{jt} = 1$$

- The pricing formula is a linear combination of J Black-Scholes formulas. It depends not only on the σ_j^2 , but also on the μ_j of the Gaussian distributions in the mixture.
- Given that, in the Gaussian (Black-Scholes) case (i.e. when $J = 1$) we have, under AAO:

$$\alpha_{bs} = -(\sigma^2)^{-1} [\mu - r + (\sigma^2/2)],$$

I.6 – Direct Modelling of Mixed-Normal Processes

Static case

- γ_j can be written as:

$$\gamma_j = \exp \left[\sigma_j^2 (\alpha - \alpha_{bs,j}) \right],$$

where α is the risk correction factor associated with the Mixed-Normal framework. Indeed, if we consider the Black-Scholes parametrization of the Gaussian distribution, with $\mu_j = m_j - \sigma_j^2/2$, we find that:

$$\alpha_{bs,j} = -\Pi_{bs,j},$$

with $\Pi_{bs,j} = (m_j - r)/\sigma_j^2$ the price of risk characterizing the j^{th} Gaussian-like market.

- We can thus rewrite γ_j in the following way :

$$\gamma_j = \exp \left[\sigma_j^2 (\Pi_{bs,j} - \Pi_{mn}) \right],$$

where $\Pi_{mn} = -\alpha$ can be interpreted as the price of risk in the Mixed-Normal market.

I.6 – Direct Modelling of Mixed-Normal Processes

Static case – Modeling extreme risks

- ▶ Particular mixture of two Gaussian distributions, one of which having a small weight associated with a strong variance. This example enables us to take into account the unlikely occurrence of an important shock on the volatility.
- ▶ We consider a geometric return y whose distribution is given by the following p.d.f :

$$f(y) = \lambda n\left(y; \mu, \frac{\sigma_1^2}{\lambda}\right) + (1 - \lambda) n\left(y; \mu, \frac{\sigma_2^2}{1 - \lambda}\right),$$

and we consider the situation in which λ tends to zero. The main characteristics of this distribution are:

$$\mathbb{E}(y) = \mu, \quad \sigma^2 = \mathbb{V}ar(y) = \sigma_1^2 + \sigma_2^2, \quad \frac{\mathbb{E}(y - \mu)^3}{\sigma^3} = 0,$$

and

$$\frac{\mathbb{E}(y - \mu)^4}{\sigma^4} = \frac{3 \left[\frac{\sigma_1^4}{\lambda} + \frac{\sigma_2^4}{1 - \lambda} \right]}{[\sigma_1^2 + \sigma_2^2]^2} \rightarrow +\infty$$
$$y \xrightarrow{D} \mathcal{N}(\mu, \sigma_2^2), \quad \text{when } \lambda \rightarrow 0.$$

I.6 – Direct Modelling of Mixed-Normal Processes

Static case – Modeling extreme risks

- ▶ So, when λ tends to 0, this distribution is in a sense close to the Gaussian distribution, but with a strong kurtosis.
- ▶ The risk neutral distribution $\mathbb{Q}$ has the following p.d.f. :

$$f^{\mathbb{Q}}(y) = \nu_1 n\left(y; \mu + \alpha \frac{\sigma_1^2}{\lambda}, \frac{\sigma_1^2}{\lambda}\right) + \nu_2 n\left(y; \mu + \alpha \frac{\sigma_2^2}{1-\lambda}, \frac{\sigma_2^2}{1-\lambda}\right)$$

where

$$\nu_1 = \frac{\lambda \exp\left(\frac{\sigma_1^2}{\lambda} \frac{\alpha^2}{2}\right)}{\lambda \exp\left(\frac{\sigma_1^2}{\lambda} \frac{\alpha^2}{2}\right) + (1-\lambda) \exp\left(\frac{\sigma_2^2}{1-\lambda} \frac{\alpha^2}{2}\right)} \quad \text{and} \quad \nu_2 = 1 - \nu_1.$$

We note that $\nu_1 \rightarrow 1$ when $\lambda \rightarrow 0$.

- ▶ The price of the European Call option can be written :

$$C_t = \nu_1 \gamma_1 C_{BS}\left(\frac{\sigma_1^2}{\lambda}, \frac{\kappa}{\gamma_1}\right) + \nu_2 \gamma_2 C_{BS}\left(\frac{\sigma_2^2}{1-\lambda}, \frac{\kappa}{\gamma_2}\right)$$

I.6 – Direct Modelling of Mixed-Normal Processes

Static case – Modeling extreme risks

- ▶ with

$$\gamma_1 = \exp\left(\mu + \alpha \frac{\sigma_1^2}{\lambda} - r + \frac{\sigma_1^2}{2\lambda}\right), \quad \gamma_2 = \exp\left(\mu + \alpha \frac{\sigma_2^2}{1-\lambda} - r + \frac{\sigma_2^2}{2(1-\lambda)}\right)$$

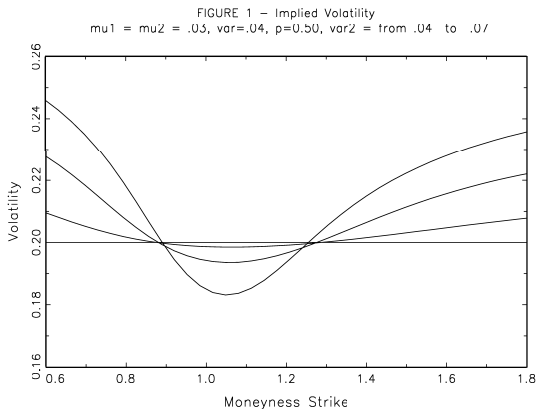
and:

$$\begin{aligned} \nu_1 \gamma_1 &\longrightarrow 1 \\ C_t &\longrightarrow 1, \quad \text{when } \lambda \rightarrow 0. \end{aligned}$$

- ▶ The first component of the historical mixture of distribution has a small weight whereas, on the contrary, it is associated with a strong weight for the risk-neutral distribution and consequently for the price of the European Call option.
- ▶ Extensions:
 - Maturity $h > 1$.
 - Non Parametric case.
 - Dynamic Mixed-Normal models.
- ▶ Applications : Smiles generated by non-normality in spite of serial independence : see next slides.

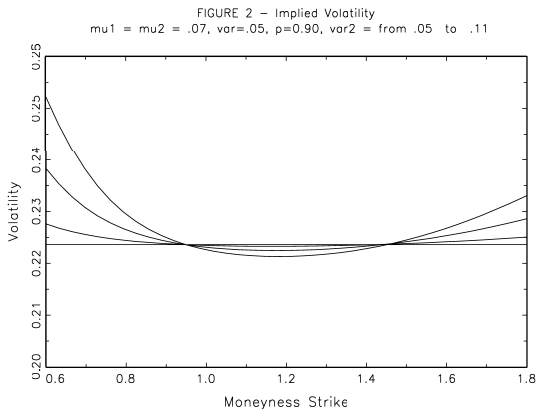
I.6 – Direct Modelling of Mixed-Normal Processes

Figures – Static case



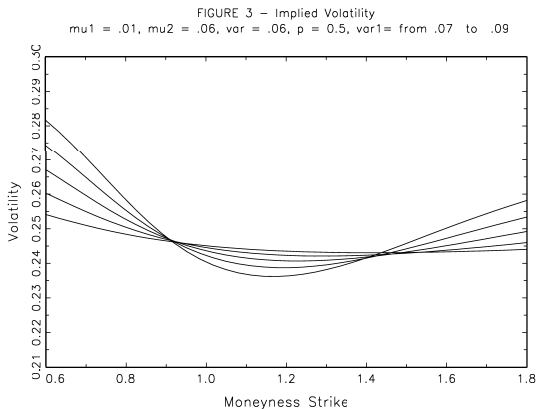
I.6 – Direct Modelling of Mixed-Normal Processes

Figures – Static case



I.6 – Direct Modelling of Mixed-Normal Processes

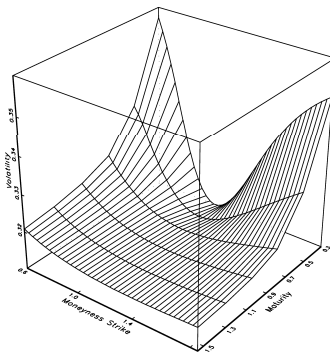
Figures – Static case



I.6 – Direct Modelling of Mixed-Normal Processes

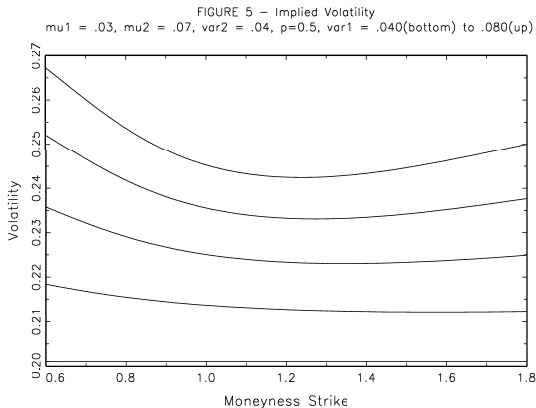
Figures – Static case

FIGURE 4 – Implied Volatility Surface
 $\mu_1=\mu_2=0.03$, $\text{var}=.025$, $\text{var}2=.035$, $p=0.5$



I.6 – Direct Modelling of Mixed-Normal Processes

Figures – Static case



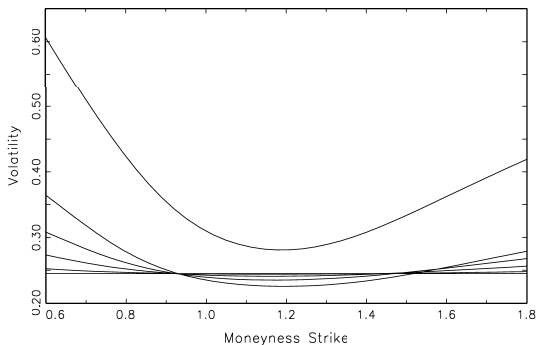
I.6 – Direct Modelling of Mixed-Normal Processes

Figures – Static case

FIGURE 6 – Implied Volatility for Extreme Risks

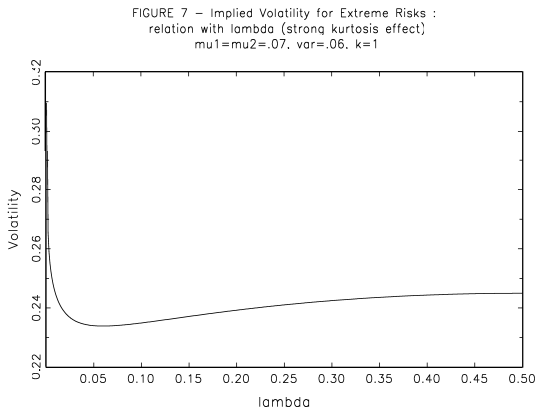
$\mu_1 = \mu_2 = 0$, $\text{var} = 0.06$

$\lambda = .0001$ (flat curve) to $.001$ (up)



I.6 – Direct Modelling of Mixed-Normal Processes

Figures – Static case



I.6 – Direct Modelling of Mixed-Normal Processes

Regime Switching case

- Here we have $K = J + 1$ and $w_{t+1} = (y_{t+1}, z'_{t+1})'$, where:

$$(y_{t+1} | \underline{y}_t, \underline{z}_t, z_{t+1} = e_j) \sim N[\mu_j(\underline{y}_t, \underline{z}_t, \theta_{11}), \sigma_j^2(\underline{y}_t, \underline{z}_t, \theta_{11})],$$

where $\mu_j(\underline{y}_t, \underline{z}_t, \theta_{11})$ and $\sigma_j^2(\underline{y}_t, \underline{z}_t, \theta_{11})$ are also denoted by μ_{jt}, σ_{jt}^2 .

We also have: $\mathbb{P}(z_{t+1} = e_j | z_t = e_i) = \pi_{ij}$.

- The historical Laplace transform and SDF are given by:

$$\varphi_t(u, v) = \sum_{j=1}^J \pi_{ij} \exp \left(v_j + u \mu_{jt} + \frac{1}{2} u^2 \sigma_{jt}^2 \right)$$

$$\mathcal{M}_{t,t+1} = \frac{\exp(-r_t + \eta_t(\underline{w}_t, \theta_2) y_{t+1} + \delta'_t(\underline{w}_t, \theta_2) z_{t+1})}{\varphi_t(\eta_t, \delta_t)}$$

- ICC: $\varphi_t(\eta_t + 1, \delta_t) = \exp(r_t) \varphi_t(\eta_t, \delta_t)$. unique solution in η_t for given δ_t .

I.6 – Direct Modelling of Mixed-Normal Processes

Regime Switching case

- The risk neutral Laplace transform is:

$$\varphi_t^{\mathbb{Q}}(u, v) = \sum_{j=1}^J \nu_{ij,t} \exp \left[v_j + u(\mu_{jt} + \eta_t \sigma_{jt}^2) + \frac{1}{2} u^2 \sigma_{jt}^2 \right]$$

with:

$$\nu_{ij,t} = \pi_{ij} \frac{\exp \left(\delta_{jt} + \eta_t \mu_{jt} + \frac{1}{2} \eta_t^2 \sigma_{jt}^2 \right)}{\sum_{j=1}^J \pi_{ij} \exp \left(\delta_{jt} + \eta_t \mu_{jt} + \frac{1}{2} \eta_t^2 \sigma_{jt}^2 \right)}$$

(we can take $\delta_{jt} = 0$).

- The $\mathbb{Q}$ -dynamics is:

$$(y_{t+1} | \underline{y}_t, \underline{z}_t, z_{t+1} = e_j) \stackrel{\mathbb{Q}}{\sim} N[\mu_{jt} + \eta_t \sigma_{jt}^2, \sigma_{jt}^2]$$

$$\mathbb{Q}(z_{t+1} = e_j | z_t = e_i, y_t) = \nu_{ij,t},$$

I.6 – Direct Modelling of Mixed-Normal Processes

Regime Switching case (Risk Premia, Market Price of Risk)

- The price at t of the European Call option with payoff $(y_{t+1} - \kappa)^+$ at $t + 1$ is :

$$C_t = \sum_{j=1}^J \nu_{ij,t} \gamma_{jt} C_{BS} \left(\sigma_{jt}^2, \frac{\kappa}{\gamma_{jt}} \right)$$

where:

$$\gamma_{jt} = \exp \left(\mu_{jt} - r_t + \eta_t \sigma_{jt}^2 + \frac{1}{2} \sigma_{jt}^2 \right)$$

- Let us consider the arithmetic risk premium :

$$\pi_{At} = \mathbb{E}_t[\exp(y_{t+1})] - \exp(r_t) = \varphi_t(1, 0) - \varphi_t^{\mathbb{Q}}(1, 0),$$

- we get, in this general switching regime case:

$$\begin{aligned} \pi_{At} &= \sum_{j=1}^J \left[\pi_{ij} \exp \left(\mu_{jt} + \frac{1}{2} \sigma_{jt}^2 \right) - \nu_{ij,t} \exp \left(\mu_{jt} + \eta_t \sigma_{jt}^2 + \frac{1}{2} \sigma_{jt}^2 \right) \right] \\ &= \sum_{j=1}^J \pi_{ij} \exp \left(\mu_{jt} + \frac{1}{2} \sigma_{jt}^2 \right) \left[1 - \frac{\exp \left(\eta_t \sigma_{jt}^2 + \eta_t \mu_{jt} + \delta_{jt} + \frac{1}{2} \eta_t^2 \sigma_{jt}^2 \right)}{\sum_{k=1}^J \pi_{ik} \exp \left(\delta_{kt} + \eta_t \mu_{kt} + \frac{1}{2} \eta_t^2 \sigma_{kt}^2 \right)} \right] \end{aligned}$$

I.6 – Direct Modelling of Mixed-Normal Processes

Regime Switching case (Risk Premia, Market Price of Risk)

- ▶ The first order expansion around $\eta_t = 0, \delta_{jt} = 0 (j = 1, \dots, J)$ (we can always assume $\delta_{Jt} = 0$) gives :

$$\pi_{At} \simeq \sum_{j=1}^J \pi_{ij} \exp(\mu_{jt} + \frac{1}{2} \sigma_{jt}^2) [-\eta_t (\sigma_{jt}^2 + \mu_{jt} - \sum_{k=1}^J \pi_{ik} \mu_{kt}) - (\delta_{jt} - \sum_{k=1}^J \pi_{ik} \delta_{kt})]$$

- ▶ In particular, if the means μ_{jt} do not depend on the regime:

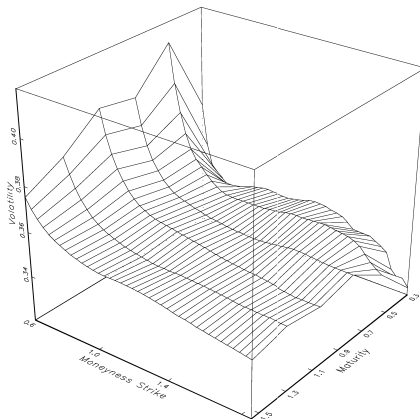
$$\begin{aligned} \pi_{At} \simeq & -\eta_t \exp(\mu_t) \sum_{j=1}^J \pi_{ij} \exp\left(\frac{1}{2} \sigma_{jt}^2\right) \sigma_{jt}^2 \\ & - \exp(\mu_t) \sum_{j=1}^J \pi_{ij} \delta_{jt} \left[\exp\left(\frac{1}{2} \sigma_{jt}^2\right) - \sum_{k=1}^J \pi_{ik} \exp\left(\frac{1}{2} \sigma_{kt}^2\right) \right] \end{aligned}$$

- ▶ It is an increasing function of $-\eta_t$, which can be seen as the Market Price of an average risk.
- ▶ π_{At} is also an increasing function of the empirical covariance between the $\exp\left(\frac{1}{2} \sigma_{jt}^2\right)$ and the $-\delta_{jt}$, $j = 1, \dots, J$, weighted by the π_{ij} ; therefore the $-\delta_{jt}$, defined up to an additive constant, can be seen as relative Market Price of risk associated to each regime.

I.6 – Direct Modelling of Mixed-Normal Processes

Figures – Regime Switching case

FIGURE 8 – Regime Switching Option Pricing Model
Implied Volatility Surface
 $\mu_1=\mu_2=.03$, $\text{var}_1=.03$, $\text{var}_2=.04$, $\pi_{11}=0.99$, $\pi_{22}=0.95$



I.7 – Direct Modelling of Conditionally Spline Dynamics

- Based on Gouriéroux and Monfort (2006). We have $K = 1$, with $w_{t+1} = y_{t+1}$, and r_{t+1} is exogenous.

- The geometric stock return follows under the historical probability a skewed Laplace distribution, namely $y_{t+1} \sim \text{ILL}(c, b_0, b_1)$, with p.d.f.:

$$\begin{aligned} f(y) &= \frac{b_0 b_1}{b_0 + b_1} \exp[b_0 (y - c)] \quad \text{if } y \leq c \\ &= \frac{b_0 b_1}{b_0 + b_1} \exp[-b_1 (y - c)] \quad \text{if } y > c, \end{aligned}$$

- The SDF is $\mathcal{M}_{t,t+1} = \frac{\exp[-r_t + \alpha_t y_{t+1}]}{\varphi(\alpha_t)}$.
- α_t is the unique solution of $\frac{\varphi(\alpha_t + 1)}{\varphi(\alpha_t)} = \exp(r_t)$.
- The risk neutral distribution ($b_0 + b_1 > 1$) is $\text{ILL}(c, b_0 + \alpha_t, b_1 - \alpha_t)$

I.7 – Direct Modelling of Conditionally Spline Dynamics

- ▶ The truncated Laplace transform of a skewed Laplace distribution can be computed explicitly (see Appendix I.D) and the price at t of a European Call option with payoff $[\exp(y_{t+1}) - \kappa]^+$ at $t + 1$ is :

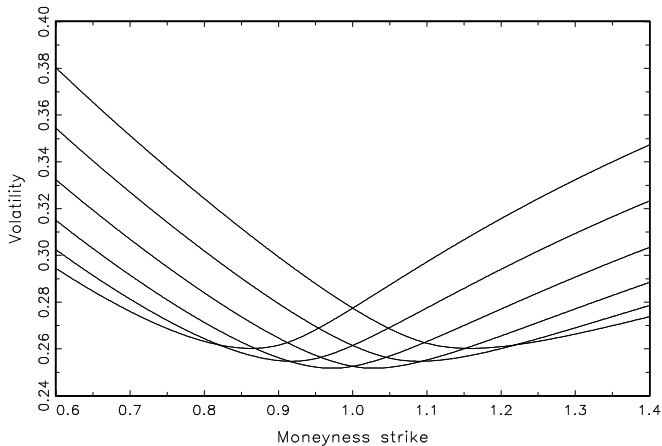
$$\begin{aligned} C_t &= \frac{b_0 + \alpha_t + 1}{(b_0 + b_1)(b_1 - \alpha_t)} \exp[-(b_1 - \alpha_t - 1)(\log \kappa - c)], \text{ if } \log \kappa \geq c \\ &= 1 - \kappa \exp(-r_{t+1}) + \frac{b_1 + \alpha_t - 1}{(b_0 + b_1)(b_0 - \alpha_t)} \\ &\quad \times \exp[(b_0 + \alpha_t - 1)(\log \kappa - c)], \text{ if } \log \kappa \leq c. \end{aligned}$$

- ▶ It depends on b_0 and b_1 , through the "tail index" $b_0 + b_1$, and on c . We have two "degrees of freedom" (one in B.S.).
- ▶ Extensions : dynamic case.
- ▶ Applications : price curves, smiles generated by asymmetry and thickness of tails (see next slides).

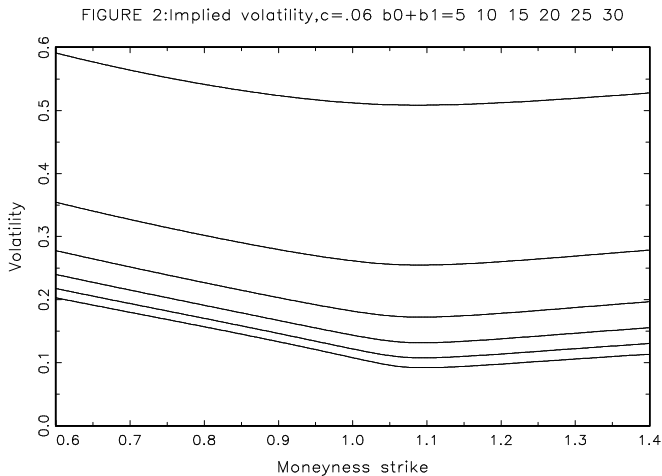
I.7 – Direct Modelling of Conditionally Spline Dynamics

Figures

FIGURE 1: Implied volatility, $b_0 + b_1 = 10$, $c = -.1$ $-.06$ $-.02$ $.02$ $.06$ $.1$

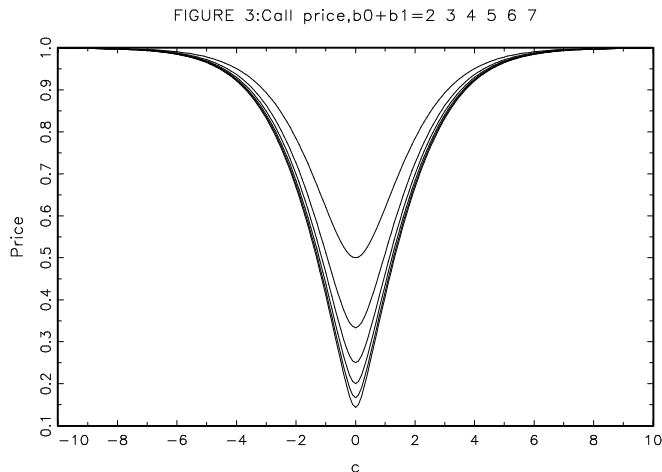


I.7 – Direct Modelling of Conditionally Spline Dynamics Figures



I.7 – Direct Modelling of Conditionally Spline Dynamics

Figures



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**Direct Modelling of
Conditionally Spline
Dynamics**

Back Modelling of Regime
Switching Dynamics

Back Modelling of
Stochastic Volatility
Dynamics.

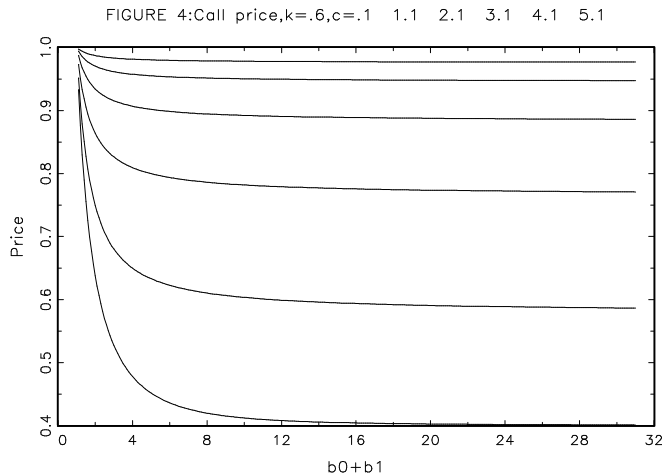
Back Modelling of Regime
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I.7 – Direct Modelling of Conditionally Spline Dynamics

Figures



I.8 – Back Modelling of Regime Switching Dynamics

□ We have $K = J + 1$ with $w_{t+1} = (y_{t+1}, z'_{t+1})'$.

► y_{t+1} is the one-period geometric stock return and z_{t+1} is the regime indicator function taking values e_j , $j = 1, \dots, J$.

► We want to specify a $\mathbb{Q}$ -affine dynamics. We assume:

$$y_{t+1} = \nu_t + \rho y_t + (\nu'_1 z_{t+1}) + (\nu'_2 z_t) + (\nu'_3 z_{t+1}) \varepsilon_{t+1}, \quad \varepsilon_{t+1} \stackrel{\mathbb{Q}}{\sim} \text{IIN}(0, 1),$$

$$\text{with } \mathbb{Q}(z_{t+1} = e_j | z_t = e_i) = \pi_{ij}. \text{ We can assume } \sum_{i=1}^J \nu_{2i} = 0.$$

► The risk-neutral conditional log-Laplace transform (given $z_t = e_i$) is:

$$\begin{aligned} \psi_{t,i}^{\mathbb{Q}}(u, v) &= \log [\mathbb{E}(\exp(uy_{t+1} + v'z_{t+1} | y_t, z_t = e_i))] \\ &= (\nu_t + \rho y_t + \nu_{2i})u + \log \sum_{j=1}^J \pi_{ij} \exp(u\nu_{1j} + \frac{1}{2}u^2\nu_{3j}^2 + v_j). \end{aligned}$$

I.8 – Back Modelling of Regime Switching Dynamics

► ICC : $\psi_{ti}^{\mathbb{Q}}(1, 0) = r_t, \forall y_t, \forall i$

$$\Leftrightarrow \nu_t + \rho y_t + \nu_{2i} + \log \sum_{j=1}^J \pi_{ij} \exp \left(\nu_{1j} + \frac{1}{2} \nu_{3j}^2 \right) = r_t, \forall y_t, \forall i$$

$$\Leftrightarrow \rho = 0, \nu_{2i} + \nu_t = r_t - \log \sum_{j=1}^J \pi_{ij} \exp \left(\nu_{1j} + \frac{1}{2} \nu_{3j}^2 \right)$$

$$\Leftrightarrow y_{t+1} = r_t + \mu(\pi, \nu_1, \nu_3)' z_t + \nu_1' z_{t+1} + (\nu_3' z_{t+1}) \varepsilon_{t+1}$$

where the i^{th} component of μ is $-\log \sum_{j=1}^J \pi_{ij} \exp \left(\nu_{1j} + \frac{1}{2} \nu_{3j}^2 \right)$.

► $\nu_{1j}, \nu_{3j}, \pi_{ij} (\pi_{ij} \geq 0, \sum_{j=1}^J \pi_{ij} = 1)$ are free parameters.

► The ICC-based risk-neutral conditional log-Laplace transform is thus:

$$\psi_{ti}^{\mathbb{Q}}(u, v) = (r_t + \mu_i) u + \log \sum_{j=1}^J \pi_{ij} \exp \left(u \nu_{1j} + \frac{1}{2} u^2 \nu_{3j}^2 + v_j \right)$$

I.8 – Back Modelling of Regime Switching Dynamics

- It is a Car(1) dynamics, with $\psi^{\mathbb{Q}}(u, v) = a'(u, v)z_t + b_t(u)$ and:

$$a_i(u, v) = \mu_i u + \log \sum_{j=1}^J \pi_{ij} \exp \left(u \nu_{1j} + \frac{1}{2} u^2 \nu_{3j}^2 + v_j \right), \quad b_t(u) = r_t u.$$

- Arbitrary market prices of risk, η_t, δ_{jt} , functions of $\underline{y}_t, \underline{z}_t$.

- The associated historical conditional Log L.T. is :

$$\begin{aligned} & \psi_{ti}(u, v) \\ &= \psi_{ti}^{\mathbb{Q}}(u - \eta_t, v - \delta_t) - \psi_{ti}^{\mathbb{Q}}(-\eta_t, -\delta_t) \\ &= (r_t + \mu_i)u \\ & \quad + \log \sum_{j=1}^J \pi_{ij} \exp \left[(u - \eta_t) \nu_{1j} + \frac{1}{2} (u - \eta_t)^2 \nu_{3j}^2 + v_j - \delta_{jt} \right] \\ & \quad - \log \sum_{j=1}^J \pi_{ij} \exp \left(-\eta_t \nu_{1j} + \frac{1}{2} \eta_t^2 \nu_{3j}^2 - \delta_{jt} \right) \\ &= (r_t + \mu_i)u + \log \sum_{j=1}^J \pi_{ijt}^* \exp \left[u(\nu_{1j} - \eta_t \nu_{3j}^2) + \frac{1}{2} u^2 \nu_{3j}^2 + v_j \right], \end{aligned}$$

I.8 – Back Modelling of Regime Switching Dynamics

$$\text{▶ with } \pi_{ijt}^* = \frac{\pi_{ij} \exp \left(-\eta_t \nu_{1j} + \frac{1}{2} \eta_t^2 \nu_{3j}^2 - \delta_{jt} \right)}{\sum_{j=1}^J \pi_{ij} \exp \left(-\eta_t \nu_{1j} + \frac{1}{2} \eta_t^2 \nu_{3j}^2 - \delta_{jt} \right)}$$

(we can take $\delta_{jt} = 0$).

- ▶ The $\mathbb{P}$ -dynamics is thus:

$$y_{t+1} = r_t + (\mu' z_t) + (\nu_1 - \eta_t \nu_3')' z_{t+1} + (\nu_3' z_{t+1}) \xi_{t+1}, \quad \xi_{t+1} \stackrel{\mathbb{P}}{\sim} \mathcal{N}(0, 1),$$

$$\text{with } \mathbb{P}(z_{t+1} = e_j | z_t = e_i) = \pi_{ijt}^*.$$

- ▶ We also have $\varepsilon_{t+1} = \xi_{t+1} + \eta_t(\nu_3' z_{t+1})$.

I.9 – Back Modelling of Stochastic Volatility Dynamics.

□ We have $K = 2$ with $w_{t+1} = (y_{t+1}, \sigma_{t+1})'$.

► y_{t+1} is the one-period geometric stock return and σ_{t+1} is the associated stochastic volatility.

► We start from the following $\mathbb{Q}$ -dynamics:

$$y_{t+1} = a_{0t} + a_1 y_t + a_2 \sigma_t^2 + a_3 \sigma_{t+1}^2 + a_4 \sigma_{t+1} \varepsilon_{t+1}$$

with $\varepsilon_{t+1} \stackrel{\mathbb{Q}}{\sim} \text{IIN}(0, 1)$, and $(\sigma_{t+1}^2 | \underline{y}_t, \underline{\sigma}_t^2) \stackrel{\mathbb{Q}}{\sim} \text{ARG}(\nu, \varphi, 1)$ (unit scale parameter).

► The associated conditional risk-neutral Laplace transform is:

$$\begin{aligned}\varphi_t^{\mathbb{Q}}(u, v) &= \mathbb{E}[\exp(uy_{t+1} + v\sigma_{t+1}^2) | \underline{y}_t, \underline{\sigma}_t^2] \\ &= \mathbb{E}\{\mathbb{E}[\exp(uy_{t+1} + v\sigma_{t+1}^2) | \underline{y}_t, \underline{\sigma}_{t+1}^2] | \underline{y}_t, \underline{\sigma}_t^2\} \\ &= \mathbb{E}_t \left\{ \exp[(a_{0t} + a_1 y_t + a_2 \sigma_t^2) u + (u a_3 + \frac{1}{2} u^2 a_4^2 + v) \sigma_{t+1}^2] \right\},\end{aligned}$$

I.9 – Back Modelling of Stochastic Volatility Dynamics.

- and we find:

$$\begin{aligned}\psi_t^{\mathbb{Q}}(u, v) &= (a_{0t} + a_1 y_t + a_2 \sigma_t^2) u + a_{\sigma}(u a_3 + \frac{1}{2} u^2 a_4^2 + v) \sigma_t^2 \\ &\quad + b_{\sigma}(u a_3 + \frac{1}{2} u^2 a_4^2 + v),\end{aligned}$$

where $a_{\sigma}(u) = \frac{\varphi u}{1-u}$ and $b_{\sigma}(u) = -\nu \log(1-u)$.

- It is a Car(1) process (with the $b(\cdot)$ function depending on time in a deterministic way). The IIC is $\psi_t^{\mathbb{Q}}(1, 0) = r_t, \forall \underline{w}_t$

$$\Leftrightarrow a_{0t} + b_{\sigma}(a_3 + \frac{1}{2} a_4^2) = r_t, \quad a_1 = 0, \quad a_{\sigma}(a_3 + \frac{1}{2} a_4^2) + a_2 = 0.$$

- We thus have:

$$y_{t+1} = r_t - b_{\sigma}(a_5) - a_{\sigma}(a_5) \sigma_t^2 + a_3 \sigma_{t+1}^2 + a_4 \sigma_{t+1} \varepsilon_{t+1}$$

with $a_5 = a_3 + \frac{1}{2} a_4^2$, $\varepsilon_{t+1} \overset{\mathbb{Q}}{\sim} \text{IIN}(0, 1)$ and

$(\sigma_{t+1}^2 | \underline{y}_t, \underline{\sigma}_t^2) \overset{\mathbb{Q}}{\sim} \text{ARG}(\nu, \varphi, 1)$. Observe there is no term in y_t .

I.9 – Back Modelling of Stochastic Volatility Dynamics.

► $\mathbb{P}$ -dynamics; we specify $\alpha_t = (\eta_t, \delta_t)'$ arbitrarily and we obtain:

$$\begin{aligned}
 \psi_t(u, v) &= \psi_t^{\mathbb{Q}}(u - \eta_t, v - \delta_t) - \psi_t^{\mathbb{Q}}(-\eta_t, -\delta_t) \\
 &= [r_t - b_{\sigma}(a_5) - a_{\sigma}(a_5) \sigma_t^2] u \\
 &\quad + a_{\sigma} \left[(u - \eta_t) a_3 + \frac{1}{2} (u - \eta_t)^2 a_4^2 + v - \delta_t \right] \sigma_t^2 \\
 &\quad - a_{\sigma} \left[-\eta_t a_3 + \frac{1}{2} \eta_t^2 a_4^2 - \delta_t \right] \sigma_t^2 \\
 &\quad + b_{\sigma} \left[(u - \eta_t) a_3 + \frac{1}{2} (u - \eta_t)^2 a_4^2 + v - \delta_t \right] \\
 &\quad - b_{\sigma} \left[-\eta_t a_3 + \frac{1}{2} \eta_t^2 a_4^2 - \delta_t \right] \\
 &= [r_t - b_{\sigma}(a_5) - a_{\sigma}(a_5) \sigma_t^2] u \\
 &\quad + a_{\sigma} \left[u (a_3 - \eta_t a_4^2) + \frac{1}{2} u^2 a_4^2 + v - \eta_t a_3 + \frac{1}{2} \eta_t^2 a_4^2 - \delta_t \right] \sigma_t^2 \\
 &\quad - a_{\sigma} \left[-\eta_t a_3 + \frac{1}{2} \eta_t^2 a_4^2 - \delta_t \right] \sigma_t^2 \\
 &\quad + b_{\sigma} \left[u (a_3 - \eta_t a_4^2) + \frac{1}{2} u^2 a_4^2 + v - \eta_t a_3 + \frac{1}{2} \eta_t^2 a_4^2 - \delta_t \right] \\
 &\quad - b_{\sigma} \left[-\eta_t a_3 + \frac{1}{2} \eta_t^2 a_4^2 - \delta_t \right]
 \end{aligned}$$

I.9 – Back Modelling of Stochastic Volatility Dynamics.

- Using the lemma of Appendix I.E we can write:

$$\begin{aligned}\psi_t(u, v) &= [r_t - b_\sigma(a_5) - a_\sigma(a_5) \sigma_t^2] u \\ &\quad + \tilde{a}_{\sigma,t} \left[u(a_3 - \eta_t a_4^2) + \frac{1}{2} u^2 a_4^2 + v \right] \sigma_t^2 \\ &\quad + \tilde{b}_{\sigma,t} \left[u(a_3 - \eta_t a_4^2) + \frac{1}{2} u^2 a_4^2 + v \right]\end{aligned}$$

where $\tilde{a}_{\sigma,t}(u) = \frac{\varphi_t u}{1 - \mu_t u}$, $\tilde{b}_{\sigma,t} = -\nu \log(1 - u \mu_t)$, and where

$$\varphi_t = \frac{\varphi}{\left(1 + \eta_t a_3 - \frac{1}{2} \eta_t^2 a_4^2 + \delta_t\right)^2} \text{ and } \mu_t = \frac{1}{1 + \eta_t a_3 - \frac{1}{2} \eta_t^2 a_4^2 + \delta_t}.$$

- We thus obtain:

$$y_{t+1} = r_t - b_\sigma(a_5) - a_\sigma(a_5) \sigma_t^2 + (a_3 - \eta_t a_4^2) \sigma_{t+1}^2 + a_4 \sigma_{t+1} \xi_{t+1}$$

$$\xi_{t+1} \stackrel{\mathbb{P}}{\sim} IIN(0, 1), \quad (\sigma_{t+1}^2 | \underline{y}_t, \underline{\sigma}_t^2) \stackrel{\mathbb{P}}{\sim} ARG(\nu, \varphi_t, \mu_t),$$

$$\varepsilon_{t+1} = \xi_{t+1} - \eta_t a_4 \sigma_{t+1}, \quad \varepsilon_{t+1} \stackrel{\mathbb{Q}}{\sim} IIN(0, 1).$$

I.10 – Back Modelling of Regime Switching GARCH Dynamics

□ We have $K = J + 1$, with $w_t = (y_t, z_t)'$.

- ▶ y_t = observable geometric return, and z_t = is a J -state Markov chain, valued in $\{e_1, \dots, e_J\}$, unobservable by the econometrician.
- ▶ We follow a Back modelling strategy and we consider specifications generalizing those proposed by Heston and Nandi (2000) [see also Elliot, Siu and Chan (2006)].
- ▶ More precisely, we develop a Regime Switching model with GARCH component (featuring a leverage effect). The $\mathbb{Q}$ -dynamics is given by:

$$y_{t+1} = \nu_t + \nu_1 y_t + \nu_2' z_t + \nu_3' z_{t+1} + \nu_4 \sigma_{t+1}^2 + \sigma_{t+1} \xi_{t+1},$$

$$\sigma_{t+1}^2 = \omega' z_t + \alpha_1 (\xi_t - \alpha_2 \sigma_t)^2 + \alpha_3 \sigma_t^2, \quad \xi_{t+1} \stackrel{\mathbb{Q}}{\sim} \text{IIN}(0, 1)$$

$$\mathbb{Q}(z_{t+1} = e_j | \underline{y}_t, \underline{z}_{t-1}, z_t = e_i) = \mathbb{Q}(z_{t+1} = e_j | z_t = e_i) = \pi_{ij}^*.$$

I.10 – Back Modelling of Regime Switching GARCH Dynamics

□ We observe that:

- a) σ_{t+1}^2 is a deterministic function of $(\xi_t, \underline{z}_t)$, and therefore of $\underline{w}_t = (\underline{y}_t, \underline{z}_t)$;
- b) in this Regime Switching GARCH(1,1) model, ξ_t replaces the usual term $\sigma_t \xi_t$;
- c) the term $\alpha_2 \sigma_t$ captures an asymmetric or "leverage" effect;

► The ICC is $\psi_t^{\mathbb{Q}}(1, 0) = r_t, \forall \underline{w}_t$, implies :

$$\begin{cases} \nu_1 = 0 \\ \nu_t + \nu'_2 \underline{z}_t + \nu_4 \sigma_{t+1}^2 = r_t - \lambda'(\nu_3, \pi^*) \underline{z}_t - \frac{1}{2} \sigma_{t+1}^2, \end{cases}$$

$$\text{with } \lambda_i(\nu_3, \pi^*) = \log \sum_{j=1}^J \pi_{ij}^* \exp(\nu_{3j}), \quad i \in \{1, \dots, J\}.$$

I.10 – Back Modelling of Regime Switching GARCH Dynamics

- The arbitrage-free $\mathbb{Q}$ -dynamics is:

$$y_{t+1} = r_t - \lambda'(\nu_3, \pi^*)z_t - \frac{1}{2}\sigma_{t+1}^2 + \nu_3'z_{t+1} + \sigma_{t+1}\xi_{t+1},$$

$$\sigma_{t+1}^2 = \omega'z_t + \alpha_1(\xi_t - \alpha_2\sigma_t)^2 + \alpha_3\sigma_t^2, \quad \xi_{t+1} \stackrel{\mathbb{Q}}{\sim} \text{IIN}(0, 1)$$

$$\mathbb{Q}(z_{t+1} = e_j | \underline{y_t}, \underline{z_{t-1}}, z_t = e_i) = \mathbb{Q}(z_{t+1} = e_j | z_t = e_i) = \pi_{ij}^*,$$

- The the $\mathbb{P}$ -dynamics is obtained by specifying $\gamma_t(\underline{w}_t, \theta_2^*)$ and $\delta_t(\underline{w}_t, \theta_2^*)$ [with, for instance, $\delta_{jt} = 0$] and in particular we have:

$$y_{t+1} = r_t - \lambda(\nu_3, \pi^*)'z_t - \frac{1}{2}\sigma_{t+1}^2 - \gamma_t(\underline{w}_t, \theta_2^*)\sigma_{t+1}^2$$

$$+ \nu_3'z_{t+1} + \sigma_{t+1}\varepsilon_{t+1},$$

$$\sigma_{t+1}^2 = \omega'z_t + \alpha_1(\xi_t - \alpha_2\sigma_t)^2 + \alpha_3\sigma_t^2,$$

$$\mathbb{P}(z_{t+1} = e_j | \underline{y_t}, \underline{z_{t-1}}, z_t = e_i) = \pi_{ij,t},$$

$$\text{with } \pi_{ij,t} = \frac{\pi_{ij}^* \exp(-\gamma_t \nu_{3j} - \delta_{jt})}{\sum_{j=1}^J \pi_{ij}^* \exp(-\gamma_t \nu_{3j} - \delta_{jt})}.$$

I.10 – Back Modelling of Regime Switching GARCH Dynamics

- Moreover, we get $\xi_{t+1} = \varepsilon_{t+1} - \gamma_t \sigma_{t+1}$ and, therefore, the equation giving σ_{t+1}^2 can be rewritten:

$$\sigma_{t+1}^2 = \omega' z_t + \alpha_1 [\varepsilon_t - (\alpha_2 + \gamma_t) \sigma_t]^2 + \alpha_3 \sigma_t^2.$$

- We observe that $w_t = (y_t, z_t')$ does not have a Car $\mathbb{Q}$ -dynamics, so the pricing seems a priori difficult. Nevertheless, the augmented factor $w_t^a := (y_t, \sigma_{t+1}^2, z_t')$ is R.N. Car and therefore the pricing methods, based on Car dynamics, apply.

- The R.N. conditional Log-Laplace transform of w_{t+1}^a , given $\underline{w}_t^a$, is :

$$\psi_t^{\mathbb{Q}}(u, v, \tilde{v}) = a_1^{\mathbb{Q}}(u, v, \tilde{v})' z_t + a_2^{\mathbb{Q}}(u, \tilde{v}) \sigma_{t+1}^2 + b_t^{\mathbb{Q}}(u, \tilde{v}),$$

where

$$a_1^{\mathbb{Q}}(u, v, \tilde{v}) = \tilde{\Lambda}(u, v, \tilde{v}, \nu_3, \omega, \pi^*) - \lambda(\nu_3, \pi^*) u$$

$$\text{with } \tilde{\Lambda}_i(u, v, \tilde{v}, \nu_3, \omega, \pi^*) = \text{Log} \sum_{j=1}^J \pi_{ij}^* \exp(u \nu_{3j} + v_j + \tilde{v} \omega_j), \quad i \in \{1, \dots, J\},$$

$$a_2^{\mathbb{Q}}(u, \tilde{v}) = -\frac{1}{2} u + \tilde{v}(\alpha_1 \alpha_2^2 + \alpha_3) + \frac{(u - 2\alpha_1 \alpha_2 \tilde{v})^2}{2(1 - 2\alpha_1 \tilde{v})}$$

$$b_t^{\mathbb{Q}}(u, \tilde{v}) = u r_{t+1} - \frac{1}{2} \text{Log}(1 - 2\alpha_1 \tilde{v}),$$

I.11 – Back Modelling of Regime Switching IG GARCH Dynamics

- We introduce, following the Back Modelling approach, several generalizations of the Inverse Gaussian (IG) GARCH model proposed by Christoffersen, Heston and Jacobs (2006).

- ▶ The strictly positive random variable y has an Inverse Gaussian (IG) distribution with parameter $\delta > 0$ [denoted $\text{IG}(\delta)$] if and only if its distribution function is given by:

$$F(y; \delta) = \int_0^y \frac{\delta}{\sqrt{2\pi z^3}} \exp[-(\sqrt{z} - \delta/\sqrt{z})^2/2] dz.$$

- ▶ The generalized Laplace transform is:

$$\mathbb{E}[\exp(\varphi y + \theta/y)] = \frac{\delta}{\sqrt{\delta^2 - 2\theta}} \exp\left(\delta - \sqrt{(\delta^2 - 2\theta)(1 - 2\varphi)}\right)$$

and $\mathbb{E}(y) = \text{Var}(y) = \delta$ [see Christoffersen, Heston and Jacobs (2006) for further details].

- ▶ First generalization: we consider switching regimes in the (historical and risk-neutral) dynamics of the geometric return y_t and in the GARCH variance σ_{t+1}^2 .

I.11 – Back Modelling of Regime Switching IG GARCH Dynamics

- ▶ Second generalization: we price not only the factor risk but also the regime-shift risk, and risk correction coefficients are in general time-varying.
- ▶ We have $K = J + 1$ with $w_t = (y_t, z_t')'$, and where z_t is the unobservable J -state homogeneous Markov chain valued in $\{e_1, \dots, e_J\}$.
- ▶ The proposed R.N. dynamics (before ICC) is given by:

$$y_{t+1} = \nu_t + \nu_1 y_t + \nu_2' z_t + \nu_3' z_{t+1} + \nu_4 \sigma_{t+1}^2 + \eta \xi_{t+1} \quad (1.4)$$

where ν_t is a deterministic function of t and

$$\xi_{t+1} | \xi_{\underline{t}}, z_{t+1} \stackrel{\mathbb{Q}}{\sim} IG \left(\frac{\sigma_{t+1}^2}{\eta^2} \right)$$

$$\sigma_{t+1}^2 = \omega' z_t + \alpha_1 \sigma_t^2 + \alpha_2 \xi_t + \alpha_3 \frac{\sigma_t^4}{\xi_t},$$

with

$$\mathbb{Q}(z_{t+1} = e_j | y_{\underline{t}}, z_{t-1}, z_t = e_i) = \mathbb{Q}(z_{t+1} = e_j | z_t = e_i) = \pi_{ij}^*.$$

I.11 – Back Modelling of Regime Switching IG GARCH Dynamics

- The R.N. conditional Log-Laplace transform of (y_{t+1}, z_{t+1}) is :

$$\begin{aligned}\psi_t^Q(u, v) &= \log \mathbb{E}_t \exp(uy_{t+1} + v'z_{t+1}) \\ &= (\nu_t + \nu_1 y_t + \nu_2' z_t + \nu_4 \sigma_{t+1}^2) u + \Lambda'(u, v, \nu_3, \pi^*) z_t \\ &\quad + \frac{\sigma_{t+1}^2}{\eta^2} \left[1 - (1 - 2u\eta)^{1/2} \right],\end{aligned}$$

where the i^{th} component of $\Lambda(u, v, \nu_3, \pi^*)$ is given by:

$$\Lambda_i(u, v, \nu_3, \pi^*) = \log \sum_{j=1}^J \pi_{ij}^* \exp(u \nu_{3j} + v_j).$$

- The ICC is $\psi_t^Q(1, 0) = r_t$, $\forall \underline{w}_t$, implying

$$r_t = \nu_t + \nu_1 y_t + \nu_2' z_t + \lambda'(\nu_3, \pi^*) z_t + \sigma_{t+1}^2 \left(\nu_4 + \frac{1}{\eta^2} \left[1 - (1 - 2\eta)^{1/2} \right] \right),$$

with the i^{th} component of $\lambda(\nu_3, \pi^*)$ given by:

$$\lambda_i(\nu_3, \pi^*) = \log \sum_{j=1}^J \pi_{ij}^* \exp(\nu_{3j}).$$

I.11 – Back Modelling of Regime Switching IG GARCH Dynamics

- Therefore, the arbitrage restriction implies:

$$\begin{cases} \nu_1 = 0, \\ \nu_2 = -\lambda(\nu_3, \pi^*), \\ \nu_4 = -\frac{1}{\eta^2} [1 - (1 - 2\eta)^{1/2}], \\ \nu_t = r_t. \end{cases}$$

- Thus, equation (I.4) becomes:

$$y_{t+1} = r_t - \lambda(\nu_3, \pi^*)' z_t - \frac{1}{\eta^2} [1 - (1 - 2\eta)^{1/2}] \sigma_{t+1}^2 + \nu_3' z_{t+1} + \eta \xi_{t+1} \quad (\text{I.5})$$

with

$$\sigma_{t+1}^2 = \omega' z_t + \alpha_1 \sigma_t^2 + \alpha_2 \xi_t + \alpha_3 \frac{\sigma_t^4}{\xi_t}$$

$$\xi_{t+1} | \underline{\xi}_t, \underline{z}_{t+1} \stackrel{\mathbb{Q}}{\sim} IG \left(\frac{\sigma_{t+1}^2}{\eta^2} \right),$$

$$\mathbb{Q}(z_{t+1} = e_j | \underline{y}_t, \underline{z}_{t-1}, z_t = e_i) = \mathbb{Q}(z_{t+1} = e_j | z_t = e_i) = \pi_{ij}^*,$$

(again, we can take $\nu_{3J} = 0$) which gives the proper R.N. dynamics (compatible with the ICC).

I.11 – Back Modelling of Regime Switching IG GARCH Dynamics

- The corresponding Log-Laplace transform is :

$$\begin{aligned} \psi_t^Q(u, v) &= \left(r_t - \lambda' z_t - \frac{1}{\eta^2} \left[1 - (1 - 2\eta)^{1/2} \right] \sigma_{t+1}^2 \right) u + \\ &\quad \Lambda'(u, v, \nu_3, \pi^*) z_t + \frac{\sigma_{t+1}^2}{\eta^2} \left[1 - (1 - 2u\eta)^{1/2} \right]. \end{aligned} \quad (I.6)$$

- Given $\gamma_t(\underline{w}_t, \theta_2^*)$ and $\delta_t(\underline{w}_t, \theta_2^*)$ (with, for instance, $\delta_{Jt} = 0$), the conditional historical Log-Laplace transform is:

$$\begin{aligned} \psi_t(u, v) &= \left(r_t - \lambda' z_t - \frac{1}{\eta^2} \left[1 - (1 - 2\eta)^{1/2} \right] \sigma_{t+1}^2 \right) u \\ &\quad + \left[\Lambda(u - \gamma_t, v - \delta_t, \nu_3, \pi^*) - \Lambda(-\gamma_t, -\delta_t, \nu_3, \pi^*) \right]' z_t \\ &\quad + \frac{\sigma_{t+1}^2}{\eta^2} \left[(1 + 2\gamma_t\eta)^{1/2} - [1 - 2(u - \gamma_t)\eta]^{1/2} \right] \\ &= \left(r_t - \lambda' z_t - \tilde{\eta}_t^{-3/2} \eta^{-1/2} \left[1 - (1 - 2\eta)^{1/2} \right] \tilde{\sigma}_{t+1}^2 \right) u \\ &\quad + \left[\Lambda(u - \gamma_t, v - \delta_t, \nu_3, \pi^*) - \Lambda(-\gamma_t, -\delta_t, \nu_3, \pi^*) \right]' z_t \\ &\quad + \frac{\tilde{\sigma}_{t+1}^2}{\tilde{\eta}_t^2} \left[1 - (1 - 2u\tilde{\eta}_t)^{1/2} \right], \end{aligned} \quad (I.7)$$

I.11 – Back Modelling of Regime Switching IG GARCH Dynamics

► with:

$$\Lambda_i(u - \gamma_t, v - \delta_t) - \Lambda_i(-\gamma_t, -\delta_t) = \log \sum_{j=1}^J \pi_{ij,t} \exp(u\nu_{3j} + v_j),$$

$$\pi_{ij,t} = \frac{\pi_{ij}^* \exp(-\gamma_t \nu_{3j} - \delta_{jt})}{\sum_{j=1}^J \pi_{ij}^* \exp(-\gamma_t \nu_{3j} - \delta_{jt})}, \quad (1.8)$$

and where $\tilde{\eta}_t = \frac{\eta}{1+2\gamma_t\eta}$ and $\tilde{\sigma}_{t+1}^2 = \sigma_{t+1}^2 \left(\frac{\tilde{\eta}_t}{\eta}\right)^{3/2}$.

► So, the non-affine historical dynamics is given by :

$$y_{t+1} = r_t - \lambda(\nu_3, \pi^*)' z_t + \nu_3' z_{t+1}$$

$$- \tilde{\eta}_t^{-3/2} \eta^{-1/2} \left[1 - (1 - 2\eta)^{1/2}\right] \tilde{\sigma}_{t+1}^2 + \tilde{\eta}_t \varepsilon_{t+1} \quad (1.9)$$

$$\varepsilon_{t+1} | \underline{\varepsilon}_t, \underline{z}_{t+1} \sim IG\left(\frac{\tilde{\sigma}_{t+1}^2}{\tilde{\eta}_t^2}\right),$$

with, using (1.5) and (1.9), $\eta \xi_{t+1} = \tilde{\eta}_t \varepsilon_{t+1}$ and

$$\tilde{\sigma}_{t+1}^2 = \tilde{\omega}_t' z_t + \tilde{\alpha}_{1,t} \tilde{\sigma}_t^2 + \tilde{\alpha}_{2,t} \varepsilon_t + \tilde{\alpha}_{3,t} \frac{\tilde{\sigma}_t^4}{\varepsilon_t}$$

$$\mathbb{P}(z_{t+1} = e_j | \underline{y}_t, \underline{z}_{t-1}, z_t = e_i) = \pi_{ij,t},$$

I.11 – Back Modelling of Regime Switching IG GARCH Dynamics

- where $\tilde{\omega}_t = \omega(\tilde{\eta}_t/\eta)^{3/2}$, $\tilde{\alpha}_{1,t} = \alpha_1(\tilde{\eta}_t/\tilde{\eta}_{t-1})^{3/2}$,
 $\tilde{\alpha}_{2,t} = \alpha_2(\tilde{\eta}_t^{3/2}\tilde{\eta}_{t-1}/\eta^{5/2})$ and $\tilde{\alpha}_{3,t} = \alpha_3\tilde{\eta}_t^{3/2}/(\tilde{\eta}_{t-1}^4\eta^{-5/2})$.
- The factor $w_{t+1} = (y_{t+1}, z'_{t+1})'$ is not a R.N. Car process, but the factor $w_{t+1}^e = (y_{t+1}, z'_{t+1}, \sigma_{t+2}^2)'$ is R.N. Car. Indeed, the R.N. conditional Log-Laplace transform of w_{t+1}^e , given $\underline{w}_t^e$, is :

$$\psi_t^{\mathbb{Q}}(u, v, \tilde{v}) = a_1^{\mathbb{Q}}(u, v, \tilde{v})' z_t + a_2^{\mathbb{Q}}(u, \tilde{v}) \sigma_{t+1}^2 + b_t^{\mathbb{Q}}(u, \tilde{v}), \quad (I.10)$$

where

$$a_1^{\mathbb{Q}}(u, v, \tilde{v}) = \tilde{\Lambda}(u, v, \tilde{v}, \nu_3, \omega, \pi^*) - \lambda(\nu_3, \pi^*) u$$

$$\text{with } \tilde{\Lambda}_i(u, v, \tilde{v}, \nu_3, \omega, \pi^*) = \log \sum_{j=1}^J \pi_{ij}^* \exp(u \nu_{3j} + v_j + \tilde{v} \omega_j), \quad i \in \{1, \dots, J\},$$

$$a_2^{\mathbb{Q}}(u, \tilde{v}) = \tilde{v} \alpha_1 - \frac{1}{\eta^2} (u [1 - (1 - 2\eta)^{1/2}] + 1 - \sqrt{(1 - 2\tilde{v} \alpha_3 \eta^4) (1 - 2(u\eta + \tilde{v} \alpha_2))})$$

$$b_t^{\mathbb{Q}}(u, \tilde{v}) = u r_{t+1} - \frac{1}{2} \log(1 - 2\tilde{v} \alpha_3 \eta^4),$$

I.12 – A Component Volatility GARCH Model

- Based on Christoffersen, Jacobs, Ornathanalai and Wang (2008, JFE).
- ▶ This paper presents a new model for the valuation of European options, in which the volatility of returns consists of two components:
 - a) a long-run component that can be modeled as fully persistent, and
 - b) a zero mean short-run component.
- ▶ Its most distinctive feature is its ability to model the volatility term structure.
- ▶ And this feature enables the component model to jointly model long-maturity and short-maturity options.
- ▶ We start from the Heston and Nandi GARCH(1, 1) model (HN (2000)), and the component model will be seen as a natural generalization.

I.12 – A Component Volatility GARCH Model

- HN (2000) propose a class of GARCH models that allows for a closed-form solution for the price of European Call options. The GARCH(1, 1) version of the model consider the following $\mathbb{P}$ -dynamics for y_{t+1} :

$$\begin{aligned}y_{t+1} &= \log(S_{t+1}/S_t) = r + \lambda \sigma_{t+1}^2 + \sigma_{t+1} \varepsilon_{t+1} \\ \sigma_{t+1}^2 &= w + b \sigma_t^2 + a(\varepsilon_t - c \sigma_t)^2,\end{aligned}\tag{I.11}$$

where ε_{t+1} is $IIN(0, 1)$, S_t is the stock price, r is the risk-free short rate (constant), λ is the market price of risk and σ_{t+1}^2 is the conditional variance for $t + 1$, known in t .

- The risk-neutral dynamics are given in HN (2000) as:

$$\begin{aligned}y_{t+1} &= r - \frac{1}{2} \sigma_{t+1}^2 + \sigma_{t+1} \varepsilon_{t+1}^* \\ \sigma_{t+1}^2 &= w + b \sigma_t^2 + a(\varepsilon_t^* - c^* \sigma_t)^2,\end{aligned}\tag{I.12}$$

where $c^* = c + \lambda + 0.5$ and $\varepsilon_{t+1}^* \sim IIN(0, 1)$ under the R.N. measure.

I.12 – A Component Volatility GARCH Model

- Given the associated moment generating function $f(t, T; \phi) = \mathbb{E}_t[\exp(\phi \ln(S_T))]$, the date- t price of a European Call option with strike price K and with maturity T is given by:

$$\begin{aligned} C(t, T, K) &= \exp(-r(T-t)) \mathbb{E}^*[\max(S_T - K, 0)] \\ &= S_t \left(\frac{1}{2} + \frac{1}{\pi} \int_0^{+\infty} \Re \left[\frac{K^{-i\phi} f^*(t, T; i\phi+1)}{i\phi S_t e^{r(T-t)}} \right] d\phi \right) \\ &\quad K \exp(-r(T-t)) \left(\frac{1}{2} + \frac{1}{\pi} \int_0^{+\infty} \Re \left[\frac{K^{-i\phi} f^*(t, T; i\phi)}{i\phi} \right] d\phi \right) \end{aligned} \quad (I.13)$$

where $f^*(t, T; i\phi)$ is the conditional characteristic function of the logarithm of the stock price under the risk-neutral measure.

- Now, by using the expression for the unconditional variance:

$$\mathbb{E}[\sigma_{t+1}^2] = \sigma^2 = \frac{w + a}{1 - b - ac^2}$$

to substitute out w , the variance process can be rewritten as:

$$\sigma_{t+1}^2 = \sigma^2 + b(\sigma_t^2 - \sigma^2) + a[(\varepsilon_t - c\sigma_t)^2 - (1 + c^2\sigma^2)] . \quad (I.14)$$

I.12 – A Component Volatility GARCH Model

- ▶ A natural generalization is to specify σ^2 as time-varying, and denoted q_{t+1} . Relation (I.14) can be generalized to:

$$\sigma_{t+1}^2 = q_{t+1} + \beta (\sigma_t^2 - q_t) + \alpha [(\varepsilon_t - \gamma_1 \sigma_t)^2 - (1 + \gamma_1^2 q_t)] . \quad (\text{I.15})$$

- ▶ Relation (I.15) can be seen as having two components: q_{t+1} is the long-run component, while $(\sigma_{t+1}^2 - q_{t+1})$ is the short-run component (with marginal mean equal to zero; see Engle and Lee (1999)).

- ▶ The model can also be written as:

$$\sigma_{t+1}^2 = q_{t+1} + \tilde{\beta} (\sigma_t^2 - q_t) + \alpha [(\varepsilon_t - \gamma_1 \sigma_t)^2 - (1 + \gamma_1^2 \sigma_t^2)] , \quad (\text{I.16})$$

where $\tilde{\beta} = \alpha \gamma_1^2 + \beta$, and we can think of:

$$\nu_{1,t} = (\varepsilon_t - \gamma_1 \sigma_t)^2 - (1 + \gamma_1^2 \sigma_t^2) = (\varepsilon_t^2 - 1) - 2 \gamma_1 \sigma_t \varepsilon_t ,$$

as a mean-zero innovation.

I.12 – A Component Volatility GARCH Model

- ▶ The model is completed by specifying the functional form of the long-run volatility component. In a first step we assume:

$$q_{t+1} = \omega + \rho q_t + \varphi [(\varepsilon_t^2 - 1) - 2\gamma_2 \sigma_t \varepsilon_t].$$

- ▶ If $\rho < 1$, we have $\mathbb{E}[q_{t+1}] = \mathbb{E}[\sigma_{t+1}^2] = \sigma^2 = \frac{\omega}{1 - \rho}$.

- ▶ We can therefore write the component volatility model as:

$$\sigma_{t+1}^2 = q_{t+1} + \tilde{\beta}(\sigma_t^2 - q_t) + \alpha \nu_{1,t} \tag{I.17}$$

$$q_{t+1} = \omega + \rho q_t + \varphi \nu_{2,t} = \sigma^2 + \rho(q_t - \sigma^2) + \varphi \nu_{2,t},$$

where $\nu_{i,t} = (\varepsilon_t^2 - 1) - 2\gamma_i \sigma_t \varepsilon_t$, $i = 1, 2$.

- ▶ In a second step, it is also considered the $\rho = 1$ case:

$$q_{t+1} = \omega + q_t + \varphi \nu_{2,t}.$$

I.12 – A Component Volatility GARCH Model

- In the case of the GARCH(1, 1) model, the multi-step forecast of the conditional variance is:

$$\mathbb{E}_t[\sigma_{t+k}^2] = \sigma^2 + \tilde{\beta}^{k-1} (\sigma_{t+1}^2 - \sigma^2),$$

where $\tilde{\beta} = b + a c^2$. It is possible now to define a convenient measure of the variance term structure for maturity T as:

$$\sigma_{t+1:t+T}^2 = \frac{1}{T} \sum_{k=1}^T \mathbb{E}_t[\sigma_{t+k}^2] = \sigma^2 + \frac{1 - \tilde{\beta}^T}{1 - \tilde{\beta}} \frac{(\sigma_{t+1}^2 - \sigma^2)}{T}.$$

- In the case of the component volatility model we have:

$$\mathbb{E}_t[\sigma_{t+k}^2 - q_{t+k}] = \tilde{\beta}^{k-1} (\sigma_{t+1}^2 - q_{t+1}),$$

$$\mathbb{E}_t[q_{t+k}] = \sigma^2 + \rho^{k-1} (q_{t+1} - \sigma^2).$$

- The associated variance term structure is:

$$\begin{aligned} \sigma_{t+1:t+T}^2 &= \frac{1}{T} \sum_{k=1}^T \mathbb{E}_t[\sigma_{t+k}^2 - q_{t+k}] + \mathbb{E}_t[q_{t+k}] \\ &= \sigma^2 + \frac{1 - \rho^T}{1 - \rho} \frac{(q_{t+1} - \sigma^2)}{T} + \frac{1 - \tilde{\beta}^T}{1 - \tilde{\beta}} \frac{(\sigma_{t+1}^2 - q_{t+1})}{T}. \end{aligned}$$

I.12 – A Component Volatility GARCH Model

- The affine risk-neutral component GARCH dynamics is:

$$\begin{aligned}\sigma_{t+1}^2 &= q_{t+1} + \tilde{\beta}^* (\sigma_t^2 - q_t) + \alpha [(\varepsilon_t^* - \gamma_1^* \sigma_t)^2 - (1 + \gamma_1^{*2} \sigma_t^2)] \\ q_{t+1} &= \omega + \rho^* q_t + \varphi [(\varepsilon_t^* - \gamma_2^* \sigma_t)^2 - (1 + \gamma_2^{*2} \sigma_t^2)],\end{aligned}\tag{I.18}$$

where:

$$\begin{aligned}\tilde{\beta}^* &= \tilde{\beta} + \alpha (\gamma_1^{*2} - \gamma_1^2) + \varphi (\gamma_2^{*2} - \gamma_2^2) \\ \rho^* &= \rho + \alpha (\gamma_1^{*2} - \gamma_1^2) + \varphi (\gamma_2^{*2} - \gamma_2^2) \\ \gamma_i^* &= \gamma_i + \lambda + 0.5.\end{aligned}\tag{I.19}$$

- The associated moment generating function $f(t, T; \phi) = \mathbb{E}_t[\exp(\phi \ln(S_T))]$ can be written as:

$$f(t, T; \phi) = S_t^\phi \exp[A_t + B_{1,t} (h_{t+1} - q_{t+1}) + B_{2,t} q_{t+1}],\tag{I.20}$$

with terminal conditions $A_T = B_{1,T} = B_{2,T} = 0$. The pricing formula (I.13) will be applied.

I.12 – A Component Volatility GARCH Model

► Notation: $\sigma_{t+1}^2 = h_{t+1}$.

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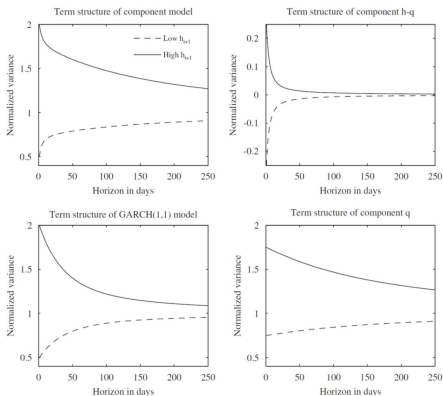


Fig. 1. Term structure of variance with high and low initial variance, component model and GARCH(1,1), normalized by unconditional variance. In the left-side panels, we plot the variance term structure implied by the component GARCH and GARCH(1,1) models for 1 through 250 days. In the right-side panels, we plot the term structure of the short-run (h – q), and long-run (q) component. The parameter values are obtained from maximum likelihood estimation on returns in Table 1. For the low initial variance (dashed lines), the initial value of q_{t+1} is set to $0.75\sigma^2$, the initial value of h_{t+1} is set to $0.5\sigma^2$, and the initial value for h_{t+1} in the GARCH(1,1) is set to $0.5\sigma^2$. For the high initial variance (solid lines), the initial value of q_{t+1} is set to $1.75\sigma^2$, the initial value of h_{t+1} is set to $2\sigma^2$, and the initial value for h_{t+1} in the GARCH(1,1) is set to $2\sigma^2$. All values are normalized by the unconditional variance σ^2 .

I.12 – A Component Volatility GARCH Model

Table 1

Maximum likelihood estimates and properties using an estimation sample of daily returns, 1962–2001

We use daily total returns from July 1, 1962 to December 31, 2001 on the Standard and Poor's 500 index to estimate the three models using maximum likelihood. Robust standard errors are calculated from the outer product of the gradient at the optimum parameter values. Persistence refers to the persistence of the conditional variance as defined in the text. Average annual volatility refers to the average annualized standard deviation during 1990–1995. Average volatility of variance refers to the average standard deviation of the conditional variance during 1990–1995. Average correlation refers to the average correlation between the return and the conditional variance during 1990–1995. Ln likelihood refers to the logarithm of the likelihood at the optimal parameter values. Option RMSE (root mean squared error) refers to the fit of the models on option prices observed during 1990–1995.

Parameter	GARCH(1,1)		Parameter	Component GARCH		Parameter	Persistent component	
	Estimate	Standard error		Estimate	Standard error		Estimate	Standard error
λ	2.231E + 00	1.123E + 00	λ	2.092E + 00	7.729E – 01	λ	2.017E – 07	4.316E – 01
w	2.101E – 17	1.120E – 07	β	6.437E – 01	2.759E – 02	β	8.822E – 01	9.931E – 03
b	9.012E – 01	4.678E – 03	α	1.580E – 06	2.430E – 07	α	2.057E – 06	1.539E – 07
a	3.317E – 06	1.380E – 07	γ_1	4.151E + 02	6.341E + 01	γ_1	2.516E + 02	2.237E + 01
c	1.276E + 02	8.347E + 00	γ_2	6.324E + 01	5.300E + 00	γ_2	1.187E + 02	1.126E + 01
			ω	8.208E – 07	7.620E – 08	ω	1.187E – 07	1.393E – 08
			φ	2.480E – 06	1.160E – 07	φ	7.966E – 07	4.599E – 08
			ρ	9.896E – 01	9.630E – 04	ρ	1.000E + 00	
Ln likelihood		33.955			34.102			34.005
Persistence		0.9553			0.9963			1.0000
Average annual volatility		0.1206			0.1174			0.1239
Average volatility of variance		7.997E – 06			1.341E – 05			1.044E – 05
Average correlation		–0.7940			–0.8849			–0.9014
Option RMSE		2.236			1.706			1.705
Normalized		1.000			0.763			0.763

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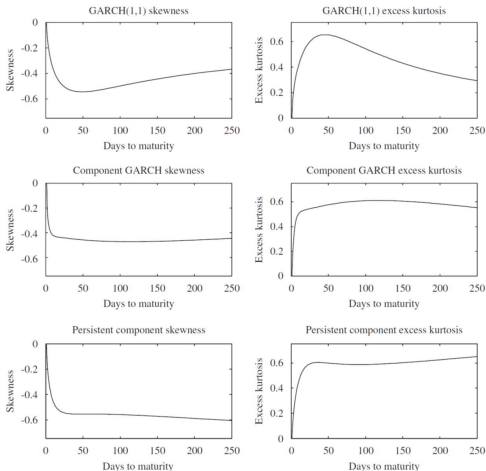


Fig. 3. Term structure of skewness and kurtosis. We use the numerical derivatives of the log moment generating function to compute the term structure of conditional skewness and kurtosis in the three GARCH models. The initial volatility is set to its long-run value in the GARCH(1,1) and component GARCH models. In the persistent component model, the initial volatility is set to the unconditional volatility from the component model. The parameter values are obtained from the maximum likelihood estimates on returns in Table 1.

- We want to compute:

$$\tilde{\varphi}_t(u; v, \gamma) = \mathbb{E}_t[\exp(u'w)\mathbb{1}_{(v'w < \gamma)}]$$

(noting $w = \tilde{w}_{t+1,T}$) knowing the function $\varphi_t(z) = \mathbb{E}_t[\exp(z'w)]$, z : complex components.

- We omit the index t for notational simplicity.
- Let us first note that, for given u and v , $\tilde{\varphi}_t(u; v, \gamma)$ is a positive increasing bounded function of γ and therefore can be seen as the c.d.f. of a positive finite measure on $\mathbb{R}$, the Fourier transform of which is:

$$\int_{\mathbb{R}} \exp(i\gamma x) d\tilde{\varphi}(u; v, \gamma) = \mathbb{E} \int_{\mathbb{R}} \exp(i\gamma x) d\tilde{\varphi}_w(u; v, \gamma),$$

where, for given w , $\tilde{\varphi}_w(u; v, \gamma)$ is the c.d.f. of the mass point $\exp(u'w)$ at $v'w$, so we get the function of x :

$$\begin{aligned} \int_{\mathbb{R}} \exp(i\gamma x) d\tilde{\varphi}(u; v, \gamma) &= \mathbb{E}[\exp(iv'w)\exp(u'w)] \\ &= \mathbb{E}[\exp(u + ivx)'w] \\ &= \varphi(u + ivx). \end{aligned}$$

► Let us now compute for any real number λ :

$$\begin{aligned}
 & A(x_0, \lambda) \\
 = & \frac{1}{2\pi} \int_{-x_0}^{x_0} \frac{\exp(i\lambda x)\varphi(u - ivx) - \exp(-i\lambda)\varphi(u + ivx)}{ix} dx \\
 = & \frac{1}{2\pi} \int_{-x_0}^{x_0} \left[\int_{\mathbb{R}} \frac{\exp[-ix(\gamma - \lambda)] - \exp[ix(\gamma - \lambda)]}{ix} d\tilde{\varphi}(u; v, \gamma) \right] dx \\
 = & \frac{1}{2\pi} \int_{\mathbb{R}} \left[\int_{-x_0}^{x_0} \frac{\exp[-ix(\gamma - \lambda)] - \exp[ix(\gamma - \lambda)]}{ix} dx \right] d\tilde{\varphi}(u; v, \gamma)
 \end{aligned}$$

► Now :

$$\begin{aligned}
 & \frac{1}{2\pi} \int_{-x_0}^{x_0} \frac{\exp[-ix(\gamma - \lambda)] - \exp[ix(\gamma - \lambda)]}{ix} dx \\
 & = \frac{-\text{sign}(\gamma - \lambda)}{\pi} \int_{-x_0}^{x_0} \frac{\sin(x | \gamma - \lambda |)}{x} dx
 \end{aligned}$$

which tends to $-\text{sign}(\gamma - \lambda)$ when $x_0 \rightarrow \infty$ (where $\text{sign}(\omega) = 1$ if $\omega > 0$, $\text{sign}(\omega) = 0$ if $\omega = 0$, $\text{sign}(\omega) = -1$ if $\omega < 0$).

Appendix I.A (Transform Analysis (cont'd))

► Therefore:

$$A(\infty, \lambda) = - \int_{\mathbb{R}} \text{sign}(\gamma - \lambda) d\tilde{\varphi}(u; v, \gamma) = -\mathbb{E} \int_{\mathbb{R}} \text{sign}(\gamma - \lambda) d\tilde{\varphi}_w(u; \theta, \gamma),$$

where $\tilde{\varphi}_w(u; v, \gamma)$ is the c.d.f. of the mass point $\exp(u'w)$ at $v'w$ and

$$\int_{\mathbb{R}} \text{sign}(\gamma - \lambda) d\tilde{\varphi}_w(u; v, \gamma) = \begin{cases} \exp(u'w) & \text{if } \lambda < v'w \\ 0 & \text{if } \lambda = v'w \\ -\exp(u'w) & \text{if } \lambda > v'w \end{cases}$$

so

$$\begin{aligned} A(\infty, \lambda) &= -\mathbb{E}[\exp(u'w)(1 - \mathbb{1}_{(v'w < \lambda)}) - \exp(u'w)\mathbb{1}_{(v'w < \lambda)}] \\ &= -\varphi(u) + 2\tilde{\varphi}(u; v, \lambda) \end{aligned}$$

and therefore:

$$\tilde{\varphi}(u; v, \gamma) = \frac{\varphi(u)}{2} + \frac{1}{2}A(\infty, \gamma)$$

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Direct Modelling of
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Back Modelling of Regime
Switching GARCH
Dynamics

Back Modelling of Regime
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Appendix I.A (Transform Analysis (cont'd))

► where:

$$\begin{aligned}\frac{1}{2}A(\infty, \gamma) &= \frac{1}{4\pi} \int_{-\infty}^{\infty} \frac{\exp(i\gamma x)\varphi(u - ivx) - \exp(-i\gamma x)\varphi(u + ivx)}{ix} dx \\ &= \frac{1}{2\pi} \int_0^{\infty} \frac{\exp(i\gamma x)\varphi(u - ivx) - \exp(-i\gamma x)\varphi(u + ivx)}{ix} dx \\ &= -\frac{1}{\pi} \int_0^{\infty} \frac{\mathcal{I}m[\exp(-i\gamma x)\varphi(u + ivx)]}{x} dx\end{aligned}$$

and finally

$$\tilde{\varphi}(u; v, \gamma) = \frac{\varphi(u)}{2} - \frac{1}{\pi} \int_0^{\infty} \frac{\mathcal{I}m[\varphi(u + ivx) \exp(-i\gamma x)]}{x} dx.$$

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Appendix I.B (Discrete-Time Girsanov (Cameron Martin) Theorem)

- If $\varepsilon_{t+1} \stackrel{\mathbb{P}}{\sim} IIN(0, Id_K)$, and under $\mathbb{Q}$ it has a conditional p.d.f. with respect to $\mathbb{P}$ given by:

$$d_{t+1}^{\mathbb{Q}}(\varepsilon_{t+1}) = \exp \left(\lambda_t' \varepsilon_{t+1} - \frac{1}{2} \lambda_t' \lambda_t \right)$$

- then, $\varepsilon_{t+1} | \underline{\varepsilon}_t \stackrel{\mathbb{Q}}{\sim} N(\lambda_t, Id_K)$, or $\varepsilon_{t+1}^{\mathbb{Q}} = \varepsilon_{t+1} - \lambda_t \stackrel{\mathbb{Q}}{\sim} IIN(0, Id_K)$.

- Proof:

$$\begin{aligned} \mathbb{E}_t^{\mathbb{Q}}[\exp(u' \varepsilon_{t+1}) | \underline{\varepsilon}_t] &= \mathbb{E}_t \left\{ \exp \left[(u + \lambda_t)' \varepsilon_{t+1} - \frac{1}{2} \lambda_t' \lambda_t \right] \right\} \\ &= \exp \left[\frac{1}{2} (u + \lambda_t)' (u + \lambda_t) - \frac{1}{2} \lambda_t' \lambda_t \right] \\ &= \exp \left[u' \lambda_t + \frac{1}{2} u' u \right]. \quad \blacksquare \end{aligned}$$

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Appendix I.C (Truncated Laplace Transform of a Normal Distribution)

► If $X \sim N(0, 1)$, then:

$$\begin{aligned}\varphi_X(u, v, \gamma) &= \mathbb{E}[\exp(uX)\mathbb{1}_{(vX < \gamma)}], \quad v > 0 \\&= \frac{1}{\sqrt{2\pi}} \int_{x < \gamma/v} \exp\left(ux - \frac{1}{2}x^2\right) dx \\&= \frac{1}{\sqrt{2\pi}} \exp\left(\frac{u^2}{2}\right) \int_{x < \gamma/v} \exp\left[-\frac{1}{2}(x-u)^2\right] dx \\&= \exp\left(\frac{u^2}{2}\right) \frac{1}{\sqrt{2\pi}} \int_{z < \gamma/v-u} \exp\left(-\frac{z^2}{2}\right) dz \\&= \exp\left(-\frac{u^2}{2}\right) \Phi\left(\frac{\gamma}{v} - u\right).\end{aligned}$$

► If $v < 0$, we change v in $-v$, X in $-X$ and we get:

$$\exp\left(\frac{u^2}{2}\right) \Phi\left(-\frac{\gamma}{v} + u\right).$$

Appendix I.D (Truncated Laplace Transform of a Skewed Laplace Distribution)

► We want to calculate $\tilde{\varphi}(u; \gamma) = \mathbb{E}[\exp(uy) \mathbb{1}_{(y < \gamma)}]$.

► If $\gamma < c$ ($u > -b_0$):

$$\begin{aligned}\tilde{\varphi}(u; \gamma) &= \exp(uc) \frac{b_0 b_1}{b_0 + b_1} \int_{-\infty}^{\gamma} \exp[u(y - c) + b_0(y - c)] dy \\ &= \exp(uc) \frac{b_0 b_1}{b_0 + b_1} \left[\frac{\exp[(b_0 + u)(y - c)]}{(b_0 + u)} \right]_{-\infty}^{\gamma} \\ &= \exp(uc) \frac{b_0 b_1}{b_0 + b_1} \frac{\exp[(b_0 + u)(\gamma - c)]}{b_0 + u},\end{aligned}$$

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Appendix I.D (Truncated Laplace Transform of a Skewed Laplace Distribution)

► If $\gamma > c$

$$\begin{aligned}
 \tilde{\varphi}(u; \gamma) &= \exp(uc) \frac{b_0 b_1}{b_0 + b_1} \int_{-\infty}^c \exp[(b_0 + u)(y - c)] dy \\
 &+ \exp(uc) \frac{b_0 b_1}{b_0 + b_1} \int_c^{\gamma} \exp[u(y - c) - b_1(y - c)] dy \\
 &= \exp(uc) \frac{b_0 b_1}{b_0 + b_1} \frac{1}{b_0 + u} \\
 &+ \exp(uc) \frac{b_0 b_1}{b_0 + b_1} \left[\frac{\exp(u - b_1)(y - c)}{u - b_1} \right]_c^{\gamma} \\
 &= \exp(uc) \frac{b_0 b_1}{b_0 + b_1} \left\{ \frac{1}{b_0 + u} + \frac{1}{u - b_1} [\exp(u - b_1)(\gamma - c) - 1] \right\}
 \end{aligned}$$

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Appendix I.E (Lemma about ARG Laplace Transform)

► If $a(u) = \frac{\varphi u}{1 - \alpha u}$, then we have:

$$\begin{aligned} a(u + \alpha; \varphi, \mu) - a(\alpha; \varphi, \mu) &= \frac{\varphi(u + \alpha)}{1 - (u + \alpha)\mu} - \frac{\varphi\alpha}{1 - \alpha\mu} \\ &= \varphi \frac{u}{(1 - \alpha\mu)^2 - u\mu(1 - \alpha\mu)} \\ &= \frac{\varphi}{(1 - \alpha\mu)^2} \frac{u}{1 - \frac{u\mu}{1 - \alpha\mu}} \\ &= \frac{\varphi^* u}{1 - u\mu^*} = a(u; \varphi^*, \mu^*), \end{aligned}$$

$$\text{where } \varphi^* = \frac{\varphi}{(1 - \alpha\mu)^2} \text{ and } \mu^* = \frac{\mu}{1 - \alpha\mu}.$$

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Appendix I.E (Lemma about ARG Laplace Transform (cont'd))

► If $b(u) = -\nu \log(1 - u\mu)$, then:

$$\begin{aligned} b(u + \alpha; \nu, \mu) - b(\alpha; \nu, \mu) &= -\nu \log[1 - (u + \alpha)\mu] + \nu \log(1 - \alpha\mu) \\ &= -\nu \log \frac{1 - (u + \alpha)\mu}{1 - \alpha\mu} \\ &= -\nu \log \left[1 - \frac{u\mu}{1 - \alpha\mu} \right] \\ &= -\nu \log(1 - u\mu^*) \\ &= b(u; \nu, \mu^*) \end{aligned}$$

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Chapter II

ASSET PRICING WITH SECOND-ORDER ESSCHER TRANSFORM

– Agenda –

- II.1 Introduction. (Slide 100)
- II.2 The Second-Order Esscher Transform. (Slide 103)
- II.3 The Exponential-Quadratic SDF modelling principle. (Slide 108)
- II.4 Conditionally Gaussian Economies. (Slide 115)
- II.5 Calibrating the Second-Order GARCH option pricing model.
(Slide 124)

II.1 – INTRODUCTION

- Our **Econometric Asset Pricing Modelling setting** is based on:
 - the **$\mathbb{P}$ -dynamics** of the factor,
 - a **SDF** $M_{t,t+1}$ (pricing the factor's risk),
 - and the **$\mathbb{Q}$ -dynamics** of the factor.

- The **purpose of the Chapter** (based on Monfort and Pegoraro (2012)) is:
 - **to introduce**, in conditionally Gaussian no-arbitrage economies **a wider bridge** between **$\mathbb{P}$** and **$\mathbb{Q}$** dynamics of **state vector** (w_t) (say), and to preserve, at the same time, its **Tractability and Flexibility**.

- **Tractability**:
 - efficient and fast MLE (under **$\mathbb{P}$**), being the likelihood function known in closed form or by standard filtering techniques;
 - explicit or quasi explicit pricing formulas (under **$\mathbb{Q}$**);

- **Flexibility**:
 - the factor (w_t) can be identified as any latent or observable (macro-financial) variable of interest;
 - non-linearities in the **$\mathbb{P}$ -dynamics** and/or in the risk-correction coefficients.

II.1 – INTRODUCTION

□ Goal achieved by:

- introducing the notion of **Exponential-Quadratic** Stochastic Discount Factor (SDF), or equivalently of **Second-Order Esscher Transform**,
- and **exploiting** one the EAPM strategies.

□ The log-pricing kernel:

- is specified as a **quadratic function of the factor** (w_t),
- and the associated sources of risk are priced by means of (possibly) non-linear stochastic first-order and second-order (say) risk-correction coefficients.

□ Focusing on security market models:

- this approach is developed in **the conditionally Gaussian framework**,
- and its usefulness is testified by the specification and calibration of what we name the **Second-Order GARCH Option Pricing Model**:
- the associated European Call option pricing formula generates a rich family of implied volatility smiles and skews able to match the typically observed ones.

II.1 – INTRODUCTION

- **With respect to the classical** exponential-affine SDF approach:
 - **we price** not only mean-based but also variance-covariance-based sources of risk,
 - and regardless an homoscedastic or conditionally heteroscedastic factors dynamics.

- **Compared with the continuous-time approach**, this methodology:
 - provides a (diffusion-like) risk-neutral conditional variance-covariance matrix of the factor **different from** the historical one,
 - **keeping** the historical and risk-neutral probabilities equivalent while, in continuous time, they would be mutually singular.

- **We deliver a generalization:**
 - of the exponential-quadratic changes of probability measures, proposed by literature on Quadratic-Gaussian (QG) or on Wishart Autoregressive (WAR) asset pricing models,
 - where, for tractability reasons, the risk-correction coefficients are constant or specified as deterministic functions of time or as affine functions of the factor.

II.2 – THE SECOND-ORDER ESSCHER TRANSFORM

- Let us introduce a **family of probability distributions**, associated with the p.d.f. f , having the (classical, First-Order) Esscher Transform as a subset.
 - This family, that we call **Second-Order Esscher Transforms**, is built upon the concept of **Second-Order Laplace Transform**.
 - It gives the possibility **TO MODIFY NOT ONLY** the mean **BUT ALSO the variance-covariance matrix** of a multivariate Gaussian distribution or the mean and the variance-covariance matrix of the components of a mixture of multivariate Gaussian distributions.
-
- **The Second-Order Laplace Transform of the p.d.f. $f(y)$ is:**

$$\varphi_S(\theta_1, \theta_2) = \int_{\mathbb{R}^n} f(y) \exp(\theta_1' y + y' \theta_2 y) d\nu(y)$$

- $\theta_1 \in \mathbb{R}^n$,
- $\theta_2 \in \mathcal{S}_n(\mathbb{R})$ an $(n \times n)$ real symmetric matrix and
- $\theta = (\theta_1, \theta_2) \in \Theta$,
- $\Theta = \{(\theta_1, \theta_2) \in \mathbb{R}^n \times \mathcal{S}_n(\mathbb{R}) : \int_{\mathbb{R}^n} f(y) \exp(\theta_1' y + y' \theta_2 y) d\nu(y) < \infty\}$
being the definition set.

II.2 – THE SECOND-ORDER ESSCHER TRANSFORM

□ **The assumption $\theta_2 \in \mathcal{S}_n(\mathbb{R})$ is not a restriction:**

- any square matrix A (say) is the sum of a symmetric matrix $(A + A')/2$ and of an antisymmetric matrix $(A - A')/2$,
- and a quadratic form associated to an antisymmetric matrix is equal to zero.

□ **The Second-Order Esscher Transform of $\mathbb{P}$ associated with (θ_1, θ_2) ,** denoted by $S_{(\theta_1, \theta_2)}(\mathbb{P})$, is given by the family of probability distributions defined by the p.d.f.:

$$g(y; \theta_1, \theta_2) = \frac{f(y) \exp(\theta_1' y + y' \theta_2 y)}{\varphi_S(\theta_1, \theta_2)}. \quad (\text{II.1})$$

- If $\theta_2 = 0$, we have First-Order Esscher Transform: denoted by $F_{(\theta_1)}(\mathbb{P})$.

II.2 – THE SECOND-ORDER ESSCHER TRANSFORM

i) **Multivariate Gaussian distribution** : $y \sim N(\mu, \Sigma)$

$$g(y; \theta_1, \theta_2) = \frac{1}{(2\pi)^{n/2} \sqrt{\det[(\Sigma^{-1} - 2\theta_2)^{-1}]} } \times \\ \exp \left[-\frac{1}{2} (y - (I - 2\Sigma\theta_2)^{-1}(\mu + \Sigma\theta_1))' (\Sigma^{-1} - 2\theta_2) (y - (I - 2\Sigma\theta_2)^{-1}(\mu + \Sigma\theta_1)) \right]$$

- We find that $g(y; \theta_1, \theta_2)$ is the p.d.f. of the family of the n -dimensional Gaussian random variable

$$N \left[(I - 2\Sigma\theta_2)^{-1}(\mu + \Sigma\theta_1), (\Sigma^{-1} - 2\theta_2)^{-1} \right]$$

if $(\Sigma^{-1} - 2\theta_2)$ is assumed to be a symmetric positive definite matrix.

- For any given (θ_1, θ_2) , the Gaussian random variable generated by (II.1) has a different mean as well as a **different variance-covariance matrix** and any n -dimensional Gaussian distribution can be reached.
- If $\theta_2 = 0$, the new probability distribution has the same variance-covariance matrix as the starting one.

II.2 – THE SECOND-ORDER ESSCHER TRANSFORM

ii) Finite Mixtures of Multivariate Gaussian distributions :

- given a finite mixture of n -dimensional Gaussian random variables with p.d.f. $f(y) = \sum_{j=1}^J \lambda_j n(y; \mu_j, \Sigma_j)$, the associated family of probability density functions generated by the Second-Order Esscher Transform is:

$$\begin{aligned} & g(y; \theta_1, \theta_2) \\ &= \sum_{j=1}^J \lambda_j^* n \left[y; (I - 2\Sigma_j\theta_2)^{-1}(\mu_j + \Sigma_j\theta_1), (\Sigma_j^{-1} - 2\theta_2)^{-1} \right], \\ & \text{with } \lambda_j^* = \frac{\lambda_j \varphi_{S,j}(\theta_1, \theta_2)}{\sum_{j=1}^J \lambda_j \varphi_{S,j}(\theta_1, \theta_2)}, \end{aligned} \tag{II.2}$$

- where the mixing weights are given by:

$$\lambda_j^* = \frac{\lambda_j \varphi_{S,j}(\theta_1, \theta_2)}{\sum_{j=1}^J \lambda_j \varphi_{S,j}(\theta_1, \theta_2)},$$

II.2 – THE SECOND-ORDER ESSCHER TRANSFORM

- where

$$\begin{aligned}\varphi_{S,j}(\theta_1, \theta_2) &= \int_{\mathbb{R}^n} \exp(\theta_1 y + y' \theta_2 y) n(y; \mu_j, \Sigma_j) dy \\ &= \exp \left[-\frac{1}{2} \log \det (I - 2\Sigma_j \theta_2) - \frac{1}{2} \mu_j' \Sigma_j^{-1} \mu_j \right. \\ &\quad \left. + \frac{1}{2} (\Sigma_j^{-1} \mu_j + \theta_1)' (\Sigma_j^{-1} - 2\theta_2)^{-1} (\Sigma_j^{-1} \mu_j + \theta_1) \right], \quad (\text{II.3}) \\ \text{and } 0 \leq \lambda_j^* \leq 1, \quad \sum_{j=1}^J \lambda_j^* &= 1.\end{aligned}$$

- This is the family of p.d.f. of a n -dimensional Finite Mixture of J Gaussian random variables:

$$N \left[(I - 2\Sigma_j \theta_2)^{-1} (\mu_j + \Sigma_j \theta_1), (\Sigma_j^{-1} - 2\theta_2)^{-1} \right],$$

$j \in \{1, \dots, J\}$, having different mean and variance-covariance matrix w.r.t. the starting one (if $\theta_2 \neq 0$), as well as different mixing weights.

- In what follows, we consider an economy between dates 0 and T . The new information in the economy at date t is denoted by w_t , while $\underline{w}_t = (w_t, w_{t-1}, \dots, w_0)$ is the entire information between 0 and t .
- The random vector w_t is called a factor or a state vector, its dimension is n and it can be made up of latent or observable variables (asset prices or macro variables).
- The historical dynamics of w_t is defined by the conditional distribution of w_{t+1} given $\underline{w}_t$, denoted by $\mathbb{P}_{t+1}$ (say) and characterized either
 - by the p.d.f. $f_t(w_{t+1}|\underline{w}_t)$
 - or the Laplace transform $\varphi_t(u|\underline{w}_t)$,
 - or the Log-Laplace transform $\psi_t(u|\underline{w}_t) = \log[\varphi_t(u|\underline{w}_t)]$.
- Assuming existence, linearity and continuity of the pricing function, and under the absence of arbitrage opportunity principle, there exists a positive Stochastic Discount Factor (SDF) $\mathcal{M}_{t,t+1}(\underline{w}_{t+1})$, for each $t \in \{0, \dots, T-1\}$, such that the price at date t of the payoff $g(\underline{w}_s)$ delivered at $s > t$ is given by:

$$p_t \left[g(\underline{w}_s) \right] = \mathbb{E}_t \left[\mathcal{M}_{t,t+1} \dots \mathcal{M}_{s-1,s} g(\underline{w}_s) \right]$$

[see Hansen and Richard (1987), Cochrane (2005) and Bertholon, Monfort and Pegoraro (2008)].

II.3 – THE EXPONENTIAL-QUADRATIC SDF MODELING PRINCIPLE

- This implies:

$$f_t^{\mathbb{Q}}(w_{t+1}|\underline{w}_t) = f_t(w_{t+1}|\underline{w}_t) d_t^{\mathbb{Q}}(w_{t+1}|\underline{w}_t),$$

$$\psi_t^{\mathbb{Q}}(u_1) = \psi_t(u_1 + \alpha_{1,t}) - \psi_t(\alpha_{1,t}), \quad u_1 \in \mathbb{R}^n.$$

- Conversely, the p.d.f. of the conditional historical distribution with respect to the R.N. one is given by :

$$\begin{aligned} d_t^{\mathbb{P}}(w_{t+1}|\underline{w}_t) &= \frac{1}{d_t^{\mathbb{Q}}(w_{t+1}|\underline{w}_t)} \\ &= \exp \left[-\alpha'_{1,t} w_{t+1} + \psi_t(\alpha_{1,t}) \right] \\ &= \exp \left[-\alpha'_{1,t} w_{t+1} - \psi_t^{\mathbb{Q}}(-\alpha_{1,t}) \right], \end{aligned} \tag{II.6}$$

since $\psi_t(u_1) = \psi_t^{\mathbb{Q}}(u_1 - \alpha_{1,t}) - \psi_t^{\mathbb{Q}}(-\alpha_{1,t})$.

II.3 – THE EXPONENTIAL-QUADRATIC SDF MODELING PRINCIPLE

□ Let us introduce now the following exponential-quadratic SDF:

$$\begin{aligned} \mathcal{M}_{t,t+1}^{(S)} &= \exp \left[-r_t(\underline{w}_t) + \alpha'_{1,t}(\underline{w}_t)w_{t+1} \right. \\ &\quad \left. + w'_{t+1}\alpha_{2,t}(\underline{w}_t)w_{t+1} - \psi_{S,t}(\alpha_{1,t}, \alpha_{2,t}|\underline{w}_t) \right], \end{aligned} \quad (11.7)$$

with:

$$\psi_{S,t}(\alpha_{1,t}, \alpha_{2,t} | \underline{w}_t) = \log \varphi_{S,t}(\alpha_{1,t}, \alpha_{2,t} | \underline{w}_t),$$

where:

$$\varphi_{S,t}(\alpha_{1,t}, \alpha_{2,t} | \underline{w}_t) = \mathbb{E}_t[\exp(\alpha'_{1,t} w_{t+1} + w'_{t+1} \alpha_{2,t} w_{t+1})],$$

is the conditional second-order Log-Laplace transform, and where $\alpha_{2,t}$ is a (time-varying) $(n \times n)$ symmetric matrix ($\alpha_{2,t} \in \mathcal{S}_n(\mathbb{R})$).

- The Risk-Neutral (R.N.) conditional distribution $\mathbb{Q}_{t+1}$ of w_{t+1} given $\underline{w}_t$, has an exponential-quadratic (in w_{t+1}) p.d.f. with respect to $\mathbb{P}_{t+1}$ given by:

$$\begin{aligned} d_t^{Q,S}(w_{t+1}|\underline{w}_t) &= \frac{\mathcal{M}_{t,t+1}^{(S)}(\underline{w}_{t+1})}{\mathbb{E}_t \left[\mathcal{M}_{t,t+1}^{(S)}(\underline{w}_{t+1}) \right]} \\ &= \exp \left[\alpha'_{1,t} w_{t+1} + w'_{t+1} \alpha_{2,t} - \psi_{S,t}(\alpha_{1,t}, \alpha_{2,t}) \right]. \end{aligned} \quad (11.8)$$

II.3 – THE EXPONENTIAL-QUADRATIC SDF MODELING PRINCIPLE

- Therefore, the R.N. conditional p.d.f. (with respect to the same measure as the corresponding conditional historical probability) is $f_t^{\mathbb{Q},S}(w_{t+1}|\underline{w}_t) = f_t(w_{t+1}|\underline{w}_t)d_t^{\mathbb{Q},S}(w_{t+1}|\underline{w}_t)$ and the R.N. conditional second-order Log-Laplace transform is:

$$\begin{aligned}\psi_{S,t}^{\mathbb{Q}}(u_1, u_2) &= \psi_{S,t}(u_1 + \alpha_{1,t}, u_2 + \alpha_{2,t}) - \psi_{S,t}(\alpha_{1,t}, \alpha_{2,t}), \\ u_1 &\in \mathbb{R}^n, \quad u_2 \in \mathcal{S}_n(\mathbb{R}).\end{aligned}\tag{II.9}$$

- Conversely, the p.d.f. of the conditional historical distribution with respect to the R.N. one is given by :

$$\begin{aligned}d_t^{\mathbb{P},S}(w_{t+1}|\underline{w}_t) &= \frac{1}{d_t^{\mathbb{Q},S}(w_{t+1}|\underline{w}_t)} \\ &= \exp \left[-\alpha'_{1,t} w_{t+1} - w'_{t+1} \alpha_{2,t} w_{t+1} + \psi_{S,t}(\alpha_{1,t}, \alpha_{2,t}) \right] \\ &= \exp \left[-\alpha'_{1,t} w_{t+1} - w'_{t+1} \alpha_{2,t} w_{t+1} - \psi_{S,t}^{\mathbb{Q}}(-\alpha_{1,t}, -\alpha_{2,t}) \right],\end{aligned}\tag{II.10}$$

since $\psi_{S,t}(u_1, u_2) = \psi_{S,t}^{\mathbb{Q}}(u_1 - \alpha_{1,t}, u_2 - \alpha_{2,t}) - \psi_{S,t}^{\mathbb{Q}}(-\alpha_{1,t}, -\alpha_{2,t})$.

II.3 – THE EXPONENTIAL-QUADRATIC SDF MODELING PRINCIPLE

Proposition 1:

- If we consider the Exponential-Quadratic Stochastic Discount Factor $\mathcal{M}_{t,t+1}^{(S)}$, the Risk-Neutral conditional distribution $\mathbb{Q}_{t+1}$ of w_{t+1} , conditionally to w_t , is the conditional Second-Order Esscher Transform of $\mathbb{P}_{t+1}$ associated with $(\alpha_{1,t}, \alpha_{2,t})$, that is:

$$\mathbb{Q}_{t+1} = S_{(\alpha_{1,t}, \alpha_{2,t})}(\mathbb{P}_{t+1}).$$

- Conversely, the historical conditional distribution $\mathbb{P}_{t+1}$ is the conditional Second-Order Esscher Transform of $\mathbb{Q}_{t+1}$ associated with $(-\alpha_{1,t}, -\alpha_{2,t})$, that is:

$$\mathbb{P}_{t+1} = S_{(-\alpha_{1,t}, -\alpha_{2,t})}(\mathbb{Q}_{t+1}).$$

II.3 – THE EXPONENTIAL-QUADRATIC SDF MODELING PRINCIPLE

- The **no-arbitrage** asset pricing setting based on an exponential-affine SDF $\mathcal{M}_{t,t+1}$, conveniently provides **explicit ICCs**, through ψ_t and $\psi_t^{\mathbb{Q}}$. These Internal Consistency Conditions (ICC) are easily extended to the case of an exponential-quadratic SDF $\mathcal{M}_{t,t+1}^{(S)}(\underline{w}_{t+1})$.
- Let us consider, for instance, the situation in which the factor w_{t+1} contains (at least) a geometric stock return and in which the short rate r_t is exogenous.
- If $w_{j,t+1} = e_j' w_{t+1}$ is a scalar geometric return (e_j being the j^{th} column of the identity matrix $I_{n \times n}$) we must have:

$$\begin{aligned}\exp(-r_t) \mathbb{E}_t^{\mathbb{Q}}[\exp(w_{j,t+1})] &= 1, \\ \iff r_t &= \psi_{S,t}^{\mathbb{Q}}(e_j, 0) \\ \iff r_t &= \psi_{S,t}(\alpha_{1,t} + e_j, \alpha_{2,t}) - \psi_{S,t}(\alpha_{1,t}, \alpha_{2,t}).\end{aligned}\tag{II.11}$$

II.4 – CONDITIONALLY GAUSSIAN ECONOMIES

Direct Modelling

- Let us assume that the factor w_t is a n -dimensional vector of geometric stock returns of risky assets, that is $w_{i,t+1} = \log(S_{i,t+1}/S_{i,t})$ for each $i \in \{1, \dots, n\}$, where $S_{i,t}$ is the price at t of asset i .
- We follow the Direct Modelling strategy [see BMP(2008)], and we thus specify first the historical dynamics ($\mathbb{P}_{t+1}$) of w_{t+1} , that is conditional normality:

$$w_{t+1} | \underline{w}_t \stackrel{\mathbb{P}}{\sim} \mathcal{N}(\mu_t, \Sigma_t), \quad (\text{II.12})$$

where we have to choose μ_t and Σ_t (VARMA models with GARCH-type noise). Second, we have to specify $\alpha_{1,t}$ and $\alpha_{2,t}$ appearing in the exponential-quadratic SDF (II.7) and to impose the ICC (II.11):

$$r_t = \psi_{S,t}(e_i + \alpha_{1,t}, \alpha_{2,t}) - \psi_{S,t}(\alpha_{1,t}, \alpha_{2,t}), \text{ where}$$

$$\begin{aligned} \psi_{S,t}(u_1, u_2) = & -\frac{1}{2} \log \det(I - 2\Sigma_t u_2) - \frac{1}{2} \mu_t' \Sigma_t^{-1} \mu_t \\ & + \frac{1}{2} (\Sigma_t^{-1} \mu_t + u_1)' (\Sigma_t^{-1} - 2u_2)^{-1} (\Sigma_t^{-1} \mu_t + u_1), \end{aligned}$$

which implies :

$$\frac{1}{2} \text{vdiag} [(\Sigma_t^{-1} - 2\alpha_{2,t})^{-1}] + (I - 2\Sigma_t \alpha_{2,t})^{-1} (\mu_t + \Sigma_t \alpha_{1,t}) = r_t e, \quad (\text{II.13})$$

where e denotes the n -dimensional unitary vector.

II.4 – CONDITIONALLY GAUSSIAN ECONOMIES

Direct Modelling

Proposition 2:

- The specification of the historical dynamics (II.12) and of the exponential-quadratic SDF (II.7) implies the following R.N. dynamics ($\mathbb{Q}_{t+1}$):

$$w_{t+1} | \underline{w}_t \stackrel{\mathbb{Q}}{\sim} N \left[(I - 2\Sigma_t \alpha_{2,t})^{-1} (\mu_t + \Sigma_t \alpha_{1,t}), (\Sigma_t^{-1} - 2\alpha_{2,t})^{-1} \right], \quad (\text{II.14})$$

that is, $\mathbb{Q}_{t+1} = S_{(\alpha_{1,t}, \alpha_{2,t})}(\mathbb{P}_{t+1})$. If we impose to (II.14) the ICC (II.13), we find that the R.N. dynamics compatible with no-arbitrage restrictions is:

$$N \left[r_t e - \frac{1}{2} v \text{diag} ((\Sigma_t^{-1} - 2\alpha_{2,t})^{-1}), (\Sigma_t^{-1} - 2\alpha_{2,t})^{-1} \right]. \quad (\text{II.15})$$

II.4 – CONDITIONALLY GAUSSIAN ECONOMIES

Direct Modelling

- This exponential-quadratic SDF change of probability measure induces **three relevant generalizations** with respect to the exponential-affine one.

- i) It provides a different R.N. conditional mean and conditional variance-covariance matrix, namely:

$$\mu_t^{\mathbb{Q}} = r_t e - \frac{1}{2} \text{vdiag}(\Sigma_t^{\mathbb{Q}}) \quad (\text{II.16})$$

$$\Sigma_t^{\mathbb{Q}} = (\Sigma_t^{-1} - 2\alpha_{2,t})^{-1},$$

because of the second-order risk-sensitivity function $\alpha_{2,t}$.

- On the contrary, in the continuous-time (Brownian motion-based) framework, the risk-neutral diffusion term has to be equal to the historical one ($\Sigma_t^{\mathbb{Q}} = \Sigma_t$, in our notation) in order to guarantee $\mathbb{Q}$ equivalent to $\mathbb{P}$, otherwise the two measures would be mutually singular.
- ii) For a given historical dynamics (estimated using stock return observations) and under no-arbitrage restrictions, the Gaussian risk-neutral dynamics, contrary to the exponential-affine setting, **still delivers free parameters** (those specifying $\alpha_{2,t}$) adapted to match derivative prices.

II.4 – CONDITIONALLY GAUSSIAN ECONOMIES

Direct Modelling

- iii) The time-varying risk-sensitivity vectors characterizing the SDF are given by :

$$\begin{aligned}\alpha_{2,t} &= \frac{1}{2} \Sigma_t^{-1} [\Sigma_t^{\mathbb{Q}} - \Sigma_t] (\Sigma_t^{\mathbb{Q}})^{-1} = \frac{\Sigma_t^{-1} - (\Sigma_t^{\mathbb{Q}})^{-1}}{2}, \text{ and} \\ \alpha_{1,t} &= (\Sigma_t^{\mathbb{Q}})^{-1} \mu_t^{\mathbb{Q}} - \Sigma_t^{-1} \mu_t,\end{aligned}\tag{II.17}$$

- and, therefore, they can be seen respectively as a normalized measure of the historical and risk-neutral **variance-covariance spread**, and as a (variance-weighted) measure of the historical and risk-neutral **mean spread**.
- In order to provide a more precise interpretation of the risk-sensitivity functions $\alpha_{1,t}$ and $\alpha_{2,t}$, let us define the scalar market risk premium between t and $t+1$, associated to any stock return $w_{i,t+1}$ for $i \in \{1, \dots, n\}$, in the following way:

$$\lambda_{t,t+1}^{(i)} = \log \mathbb{E}_t[\exp(w_{i,t+1})] - r_t,$$

and let us collect them in the vector $\lambda_{t,t+1} = \left[\lambda_{t,t+1}^{(1)}, \dots, \lambda_{t,t+1}^{(n)} \right]'$.

II.4 – CONDITIONALLY GAUSSIAN ECONOMIES

Direct Modelling

- Then, from (II.13), we can write the following n -dimensional generalized market risk premium (*GMRP*):

$$\begin{aligned}\lambda_{t,t+1} &= \mu_t + \frac{1}{2} \text{vdiag} \Sigma_t - r_t e \\ &= \mu_t - \mu_t^{\mathbb{Q}}(\alpha_{2,t}) - \frac{1}{2} \text{vdiag} (\Sigma_t^{\mathbb{Q}}(\alpha_{2,t}) - \Sigma_t) \\ &= \lambda_{t,t+1}^F + [\mu_t^{\mathbb{Q}}(0) - \mu_t^{\mathbb{Q}}(\alpha_{2,t})] - \frac{1}{2} \text{vdiag} (\Sigma_t^{\mathbb{Q}}(\alpha_{2,t}) - \Sigma_t),\end{aligned}\tag{II.18}$$

- with:

$$\lambda_{t,t+1}^F := (\mu_t - \mu_t^{\mathbb{Q}}(0)) = -\Sigma_t \alpha_{1,t}$$

denoting the n -dimensional (first-order) risk premium associated to an exponential-affine SDF (and e denoting the n -dimensional unitary vector).

- Relation (II.18) shows the role played by $\alpha_{2,t}$, that is, the consequences on the risk premium played by the introduction of a quadratic term in the SDF.

Direct Modelling

Lecture 6

Monfort &
Pegoraro & Renne

The Second-Order Esscher Transform

The Exponential-Quadratic SDF modelling principle

Conditionally Gaussian Economies

Calibrating the Second-Order GARCH option pricing model

- If we assume $\alpha_{2,t} = 0$ (exponential-affine SDF), we find:

$$\lambda_{t,t+1} = \lambda_{t,t+1}^F;$$

- the risk premium is determined *comparing only* μ_t and $\mu_t^{\mathbb{Q}}$, and $-\alpha_{1,t}$ can be interpreted as a first moment-based risk premium per unit of conditional variance-covariance.
- If $\alpha_{2,t} \neq 0$:
 - then $\lambda_{t,t+1}$ depends positively on the mean spread $(\mu_t - \mu_t^{\mathbb{Q}}(\alpha_{2,t}))$,
 - and negatively on the variance-covariance spread $\text{vdiag}(\Sigma_t^{\mathbb{Q}}(\alpha_{2,t}) - \Sigma_t)$;
 - the empirical evidence finds to be (both) positive [see, among the others, Bakshi and Madan (2006)].
- If we consider the particular scalar ($n = 1$) static case ($r_{t+1} = r, \sigma_t = \sigma, \alpha_{2,t} = \alpha_2$), we find the following explicit *Second-Order Black and Scholes pricing formula* (for European Call options):

$$C_{BS}^{(S)}(t, h; K, S_t, r, \sigma^2, \alpha_2) = C_{BS}(t, h; K, S_t, r, (\sigma^Q)^2(\alpha_2)), \quad (\text{II.19})$$

in which α_2 is an additional parameter with respect to the classical Black and Scholes one ($\alpha_2 = 0$ implies $C_{BS}^{(S)}(t, h; K, S_t, r, \sigma^2, 0) = C_{BS}(t, h; K, S_t, r, \sigma^2)$).

II.4 – CONDITIONALLY GAUSSIAN ECONOMIES

Back Modelling

- This previously mentioned source of flexibility can be further exploited by specifying $\alpha_{2,t}$ as a deterministic function of time, still leading to an explicit pricing formula.
- Moreover, we can easily propose, in a dynamic setting, richer Call option pricing formulas if we assume σ_t^2 and $\alpha_{2,t}$ functions of the date t information.
- Let us now adopt the Back Modelling strategy opening the way for a *tractable and flexible* specification of the asset pricing model of interest.
- Let us assume that the R.N. dynamics ($\mathbb{Q}_{t+1}$) of w_{t+1} is given by:

$$w_{t+1} | \underline{w}_t \stackrel{\mathbb{Q}}{\sim} N(\mu_t^{\mathbb{Q}}, \Sigma_t^{\mathbb{Q}}), \quad (II.20)$$

with the associated conditional second-order Log-Laplace transform

$$\begin{aligned} \psi_{S,t}^{\mathbb{Q}}(u_1, u_2) \\ = -\frac{1}{2} \log \det (I - 2\Sigma_t^{\mathbb{Q}} u_2) - \frac{1}{2} \mu_t^{\mathbb{Q}'} (\Sigma_t^{\mathbb{Q}})^{-1} \mu_t^{\mathbb{Q}} \\ + \frac{1}{2} [(\Sigma_t^{\mathbb{Q}})^{-1} \mu_t^{\mathbb{Q}} + u_1]' [(\Sigma_t^{\mathbb{Q}})^{-1} - 2u_2]^{-1} [(\Sigma_t^{\mathbb{Q}})^{-1} \mu_t^{\mathbb{Q}} + u_1], \end{aligned} \quad (II.21)$$

II.4 – CONDITIONALLY GAUSSIAN ECONOMIES

Back Modelling

- and we impose the ICC $\psi_{S,t}^{\mathbb{Q}}(e_i, 0) = r_t$ for all/ $i \in \{1, \dots, n\}$, that is:

$$\begin{aligned} r_t &= -\frac{1}{2} \mu_t^{\mathbb{Q}'} (\Sigma_t^{\mathbb{Q}})^{-1} \mu_t^{\mathbb{Q}} + \frac{1}{2} [(\Sigma_t^{\mathbb{Q}})^{-1} \mu_t^{\mathbb{Q}} + e_i]' \Sigma_t^{\mathbb{Q}} [(\Sigma_t^{\mathbb{Q}})^{-1} \mu_t^{\mathbb{Q}} + e_i] \\ &= \frac{1}{2} e_i' \Sigma_t^{\mathbb{Q}} e_i + e_i' \mu_t^{\mathbb{Q}} \quad \forall i \in \{1, \dots, n\}. \end{aligned} \tag{II.22}$$

- From (II.22) we have $\mu_t^{\mathbb{Q}} = r_t e - \frac{1}{2} v \text{diag} \Sigma_t^{\mathbb{Q}}$ and, therefore, we find the (no-arbitrage) risk-neutral dynamics:

$$N \left[r_t e - \frac{1}{2} v \text{diag} \Sigma_t^{\mathbb{Q}}, \Sigma_t^{\mathbb{Q}} \right], \tag{II.23}$$

which is easily made affine (tractability) simply assuming, for instance, $\Sigma_t^{\mathbb{Q}} = \Sigma^{\mathbb{Q}}$ and without making any assumption or imposing any restriction on the risk-correction coefficients.

II.4 – CONDITIONALLY GAUSSIAN ECONOMIES

Back Modelling

Proposition 3:

- The historical dynamics $\mathbb{P}_{t+1}$ is given, for any non-linear stochastic risk-correction coefficients $\alpha_{1,t}$ and $\alpha_{2,t}$, by $\mathbb{P}_{t+1} = S_{(-\alpha_{1,t}, -\alpha_{2,t})}(\mathbb{Q}_{t+1})$ and we have:

$$w_{t+1} | \underline{w}_t \stackrel{\mathbb{P}}{\sim} N(\mu_t, \Sigma_t),$$

$$\mu_t = (I + 2\Sigma_t^{\mathbb{Q}}\alpha_{2,t})^{-1} \left(r_t e - \frac{1}{2} v \text{diag} \Sigma_t^{\mathbb{Q}} - \Sigma_t^{\mathbb{Q}} \alpha_{1,t} \right), \quad (\text{II.24})$$

$$\Sigma_t = ((\Sigma_t^{\mathbb{Q}})^{-1} + 2\alpha_{2,t})^{-1}.$$

- For any given R.N. conditionally Gaussian dynamics (possibly affine) and for any $\alpha_{1,t}$ and $\alpha_{2,t}$:
- i) historical dynamics is still conditionally Gaussian and
 - ii) any conditional mean and any conditional variance-covariance matrix can be reached thanks to the **non-linearities** of the risk-correction coefficients.

II.5 – SECOND-ORDER GARCH OPTION PRICING MODEL

- Let us assume that the one-period geometric stock return $w_{t+1} = y_{t+1} = \log(S_{t+1}/S_t)$, where S_t is the price at date t of the risky asset, is characterized, conditionally to $\underline{y}_t$, by the following historical distribution ($\mathbb{P}_{t+1}$, say):

$$\begin{aligned}y_{t+1} &= \mu_0 + \mu_1 y_t + \sigma_{t+1} \varepsilon_{t+1}, \\ \sigma_{t+1}^2 &= \omega_0 + \omega_1 (\sigma_t \varepsilon_t - \omega_2)^2 + \omega_3 \sigma_t^2, \\ \varepsilon_{t+1} | \underline{\varepsilon}_t &\stackrel{\mathbb{P}}{\sim} N(0, 1).\end{aligned}\tag{II.25}$$

that is, y_{t+1} follows a AR(1)-GARCH(1,1) process with leverage effect.

- From Proposition 2, we easily obtain that the no-arbitrage risk-neutral conditional distribution $\mathbb{Q}_{t+1} = S_{(\alpha_{1,t}, \alpha_{2,t})}(\mathbb{P}_{t+1})$ of y_{t+1} , conditionally to $\underline{y}_t$, is given by:

$$\begin{aligned}y_{t+1} &= r_t - \frac{1}{2} (\sigma_{t+1}^{\mathbb{Q}})^2 + \sigma_{t+1}^{\mathbb{Q}} \xi_{t+1}, \\ (\sigma_{t+1}^{\mathbb{Q}})^2 &= \frac{\sigma_{t+1}^2}{(1 - 2\sigma_{t+1}^2 \alpha_{2,t})}, \quad \xi_{t+1} | \underline{\xi}_t \stackrel{\mathbb{Q}}{\sim} N(0, 1).\end{aligned}\tag{II.26}$$

- If $\alpha_{2,t} = 0$, we have $(\sigma_{t+1}^{\mathbb{Q}})^2 = (\sigma_{t+1})^2$ and the risk-neutral dynamics is **completely specified** by historical parameters only.

II.5 – SECOND-ORDER GARCH OPTION PRICING MODEL

- The price at date t of a European Call option with underlying stock price S_t , residual maturity τ and moneyness strike $\kappa_t = \frac{K}{S_t}$ is given, assuming an exogenous short rate, by :

$$\begin{aligned} C(t, \tau; \kappa_t) &= \exp\left(-\sum_{i=1}^{\tau} r_{t+i-1}\right) \mathbb{E}_t^{\mathbb{Q}}[S_{t+\tau} - K]^+, \\ &= \exp\left(-\sum_{i=1}^{\tau} r_{t+i-1}\right) S_t \mathbb{E}_t^{\mathbb{Q}}[\exp(y_{t+1} + \dots + y_{t+\tau}) - \kappa_t]^+, \\ &\approx \exp\left(-\sum_{i=1}^{\tau} r_{t+i-1}\right) S_t \frac{1}{S} \sum_{s=1}^S [\exp(y_{t+1}^{(s)} + \dots + y_{t+\tau}^{(s)}) - \kappa_t]^+, \end{aligned} \tag{II.27}$$

- where, for each $s \in \{1, \dots, S\}$, the values $(y_{t+1}^{(s)}, \dots, y_{t+\tau}^{(s)})$ entering in the pricing formula are simulated from the no-arbitrage risk-neutral conditional distribution (II.26).
- The implementation of the Second-Order GARCH Option Pricing Model is based on the following steps.

II.5 – SECOND-ORDER GARCH OPTION PRICING MODEL

- iii) Given this estimator $\hat{\theta}_{\mathbb{P}}$, we **calibrate the risk-neutral parameters** $\theta_{\mathbb{Q}} = (\beta_0, \beta_1, \beta_2)'$ characterizing φ_t (i.e. α_{2t}):
- in order to study the larger set of implied volatility shapes that our model is able to provide, compared to the one provided by the classical GARCH(1,1) case with $\alpha_{2t} = 0$ (our benchmark).
 - Indeed, for any given $\hat{\theta}_{\mathbb{P}}$, if $\alpha_{2t} = 0$: $C(t, \tau; \kappa_t) = f(t, \tau; \kappa_t; \hat{\theta}_{\mathbb{P}})$ and, thus, the implied volatility curve (smile or smirk) turns out to be a function of $\hat{\theta}_{\mathbb{P}}$ only.
- Geometric stock returns are simulated from the risk-neutral dynamics (II.26) adopting the empirical martingale simulation method of Duan and Simonato (1998).
- We have assumed a constant 5% annual short rate and $S = 10000$ simulations.
 - The Black and Scholes (1973) implied volatilities $\sigma_{IV,t} = \sigma_{IV,t}(\kappa_t, \tau; \alpha_{2t})$ (say) provided by our calibration exercise for an European Call option with $S_t = 1$, $\kappa_t \in (0.8, 1.2)$ and $\tau = 30$ days, are shown in the next figures.

II.5 – SECOND-ORDER GARCH OPTION PRICING MODEL

Next four figures shows $\sigma_{IV,t}$ obtained by assuming in (II.28):

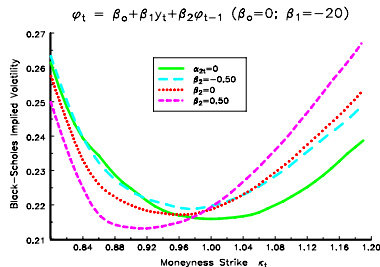
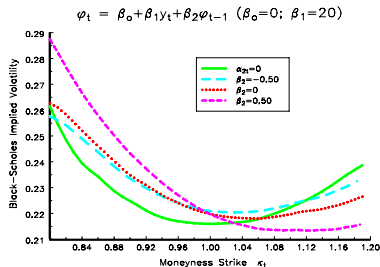
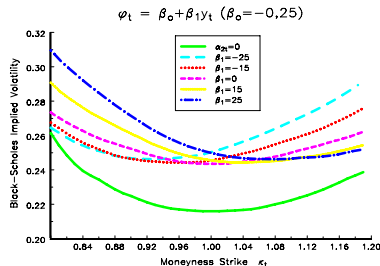
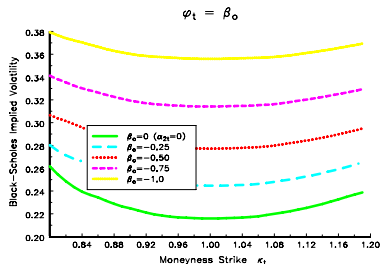
□ **Top panels:**

- $\varphi_t = \beta_0$, with $\beta_0 \in \{0, -0.25, -0.50, -0.75\}$ (left panel)
- and $\varphi_t = \beta_0 + \beta_1 y_t$, with $\beta_0 = -0.25$ and $\beta_1 \in \{-25, 0, 25\}$ (right panel).
- Our option pricing formula is potentially able to match implied volatility smiles and smirks with different degrees of curvature and steepness at different (larger) levels than the benchmark case $\alpha_{2t} = 0$.

□ **Bottom panels (introducing the lagged term $\beta_2 \varphi_{t-1}$):**

- $\varphi_t = \beta_0 + \beta_1 y_t + \beta_2 \varphi_{t-1}$, with $\beta_0 = 0$, $\beta_2 \in \{-0.75, 0, 0.50\}$ and $\beta_1 = 20$ (left panel),
- while $\beta_1 = -20$ (right panel).

II.5 – SECOND-ORDER GARCH OPTION PRICING MODEL



II.5 – SECOND-ORDER GARCH OPTION PRICING MODEL

- This example shows the **usefulness that the discrete-time setting** may provide in the specification of an asset pricing model:
 - **more general change of probability measure**
 - *and* a **more flexible specification** of the market price risk.
 - Both features are shown to be relevant.

- Next two figures shows:
 - smiles and skews with different degrees of steepness and curvatures
 - at larger levels of volatility.

II.5 – SECOND-ORDER GARCH OPTION PRICING MODEL

